

Explainable Ensemble Learning for Global, Multi-Class, Forward-in-Time Seismic Event Discrimination

Jamie Swaine

Department of Earth Science and Engineering, Imperial College London, London, SW7 2AZ, UK.

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Verification of the Comprehensive Nuclear-Test-Ban Treaty (CTBT) requires distinguishing nuclear explosions from other seismic phenomena, where a false-negative is geopolitically catastrophic. Published discriminators report within-region accuracies of 0.95–0.99 but show poor spatial transfer, suggesting they encode path, site and catalogue information rather than source-physics. We test that hypothesis directly. Five archives are unified into 436,673 globally distributed traces across six classes, processed and fed to six architecturally diverse classifiers – five implemented from literature and one original – under a forward-in-time event-level split protocol. Individual model outputs are adjusted for confidence via calibration (best performer: **SWP-Net** macro-F1 0.88). Stacking calibrated outputs via meta-learner reaches trace-level macro-F1 0.91. The ensemble is characterised under pick-perturbation, training-volume reduction, and spatial transfer experiments. Event geometry is deliberately withheld, a decision validated by a depth-binning experiment. Self-supervised latent features are provided to the meta-learner alongside hand-crafted ones and are also useful for high-volume class triage. Explainability methods (saliency, integrated, gradients, Grad-CAM, attention rollout, time-occlusion) show attribution concentration on salient seismological features – evidence the models learn source-physics. Where models fail to transfer spatially, the cause proves more archival than physical, a risk a heterogeneous training corpus is shown to mitigate. This work showcases the potential for a trustworthy and interpretable global multi-class seismic event discrimination system.

I. INTRODUCTION

Seismic networks record ground motion continuously, but deciding what generates a detected signal is a challenging problem. Classical discrimination methods exploit differences in source mechanism: explosions are compressional, shallow and have an impulsive onset, resulting in reduced shear-wave energy compared to earthquakes.[1, 2] These features motivate the use of discriminants such as body to surface wave magnitude ratio (m_b/M_s), P/S amplitude ratios, focal depth, and first-motion polarity which have underpinned treaty verification for decades.[3–6] These discriminants have clear limitations: each requires a minimum signal-to-noise ratio, focal depth is not well-constrained – trading off against origin time and epicentral location – and ratios are calibrated regionally meaning that a m_b/M_s threshold tuned for one crustal regime may not transfer well to another.[7–9] The events for which discrimination is most difficult are small, sparsely recorded and atypical – precisely the events that a low-yield evasive test would resemble.[10, 11]

The cost of this problem is asymmetric. A false positive costs analyst time, whereas a false negative of a nuclear test can have catastrophic geopolitical consequences. Machine learning is attractive here for two reasons. The first is that seismic networks generate far more detections than analysts can examine and a discriminant that requires an analyst to manually pick phases and consider the crustal region does not scale to this high volume; events are therefore triaged and small sparsely recorded events may go unexamined. Cheap per-trace screening raises the number of candidates that receive atten-

tion, and therefore elevates detected true positives.[12–14] The second concerns representation. Classical discriminants are hand-crafted to be interpretable not necessarily to be optimally discriminative. A model trained on a high-volume of traces may find separating structure that isn’t obviously visible to humans, and Sec. III A shows this concretely: deep members lead classical members by 0.26 macro-F1 on the same traces. The discrimination problem prevails beyond treaty verification and is shared by induced seismicity, mining and natural hazard monitoring.[15] In all of them path and site effects can dominate source character, instrumentation is heterogeneous across time and space so archive and regional provenance are themselves learnable signals, and the rarest classes are often of most interest.[16, 17]

The five studies forming this project’s literature review span four model families. Barama *et al.* 2023 train a 1-D CNN directly on raw vertical-component waveforms, after assembling the GTUNE archive of underground nuclear explosions (Barama *et al.* 2022) that this work builds on, reporting 95.6–99.6% accuracy on 3-class discrimination globally.[18, 19] Kasburg *et al.* 2025 apply a vision transformer to spectrogram patches and report within-region accuracy up to 100% on 2-class discrimination; they are also the only group in the set to interrogate attribution or spatial transfer.[20] Kasburg’s own conclusion is that a global model spanning a wider range of event types remains to be built. Kong *et al.* 2022 fuse a waveform CNN with a parallel branch of scalar physics descriptors, arguing that learned and physically-motivated representations both carry necessary information, reporting 0.99 AUC within-region on 2-class discrimination.[21] Li *et al.* 2024 and Pignatelli *et al.*

2025 feed hand-crafted spectral and temporal features to an XGBoost and an RBF SVM respectively both reporting >95% accuracy on a 2-class problem.[22, 23] These methods report comparable or better within-region separation at an inference cost of milliseconds.[20] None of them, however, confidently demonstrate that models survive leaving the region they were trained in, and all use random splits, reporting interpolation accuracy (which may inflate performance claims) rather than addressing the forward-in-time inference a monitoring system performs.[24–26]

These limitations yield a central question: does a multi-class discriminator learn source physics, or does it learn spurious archive, path and site effects? The distinction is invisible to a within-region test split and vital for treaty verification, where a model is applied at sites it was not trained on. We decompose this into four sub-questions. **Q1**: can a single pipeline show strong multi-class performance on a globally heterogeneous corpus under a forward-in-time split, rather than the within-region interpolation? **Q2**: where catalogue metadata offers a shortcut, does the system take it, and can that be detected? **Q3**: does the region a model was trained in determine where it can be applied, and where performance fails to transfer, is the cause physical or archival? **Q4**: where does each member’s evidence actually sit? On the source, or on artefacts of windowing, padding or the parent archive?

We retrain and tune (App. G) six architecturally diverse classifiers – five reimplemented from the studies above, one original (**SWP-Net**, fusing the two strongest reviewed designs) – on one 436,673-trace six-class corpus through a single pipeline, then calibrating and stacking them under a meta-learner and scoring at both trace and event level (**Q1**). We then deliberately reinstate the geometry we had withheld, and watch the headline metrics inflate (Sec. III C, **Q2**). We transfer models between three disjoint longitude sectors and forward in time to an unseen test site (Secs. III D, III E, **Q3**). We stress-test the two inputs a monitoring system cannot guarantee, pick accuracy and training volume (Secs. III F, III G, **Q1**, **Q4**). Finally we assess attribution across five XAI (eXplainable AI) methods matched to each model family and investigate the impact of pick perturbation on prediction outcomes (Sec. III I, Sec. III F, **Q4**). A system that faithfully learns source-physics should be robust to all these scenarios.

II. METHODOLOGY

A. Data and Preprocessing

1. Archive unification

Five archives were unified into a single vertical-component globally stretching dataset. These were: GTUNE (Georgia Tech Underground Nu-

clear Explosions)[18]; STEAD (Stanford Earthquake Dataset), contributing its noise class only[27]; PNW (Pacific Northwest AI-ready dataset) yielding earthquakes and explosions[28]; Incorporated research institute for seismology/U.S geological survey (IRIS/USGS) providing earthquakes, deep earthquakes, explosions and volcanic eruptions[29–33]; and AWE (Atomic Weapons Establishment) Blacknest chemical explosions.[34] The unified set comprises 436,673 traces. We unify under GTUNE convention as it contains the most important class – nuclear explosions.

2. Forward-in-time generalisation

Traces are split into 297,194 train, 66,679 validation and 72,800 test. Splitting is performed at the event level, so all traces of one physical event are assigned together, preventing leakage across the boundary and within each class the split is chronological, so test events postdate training events. This is the only protocol reflecting operational use, and therefore the regime under which all experiments in this work are performed. A leakage audit was performed by grouping traces within a spatial and temporal tolerance (0.5° or 55.6 km and 5 seconds) and checking that no group was split across the train/val/test boundary. No leakage was found – an important condition for **Q1**. [24]

3. Preprocessing

The pipeline is: demean $\rightarrow$ resample to 20 sps $\rightarrow$ zero-phase butterworth bandpass 0.5–9 Hz $\rightarrow$ pick-anchored 60 s window $\rightarrow$ SNR (signal-to-noise ratio) filter $\rightarrow$ per-trace z-score normalisation.

Demeaning removes DC offset, resampling to 20 sps follows GTUNE’s native rate (lower-rate traces were omitted), the 0.5 Hz lower corner removes the oceanic microseism (~ 0.1 – 0.3 Hz)[35, 36], and the 9 Hz upper corner sits below Nyquist (10 Hz) with headroom for roll-off. Filtering is zero-phase, so no displacement of the P-arrival relative to its pick occurs.[37]

The 60 s window is anchored on the P-arrival at 20 s pre-pick and 40 s post-pick, placing the pick at sample 400 of 1200. Anchoring prevents the arrival position becoming an archive specific shortcut (**Q4**). [16] SNR ratio is computed as the ratio of the sum of the square of the points in a post-pick signal window (40 s) with those of the pre-pick window (20 s). Traces whose SNR ratio falls below 3.0 are discarded, with `noise` and `volcanic_eruption` exempt since the former is non-impulsive and background magma motion obfuscate the latter’s impulsive onset.[38] Per-trace z-scoring follows, with the consequence that absolute amplitude is not comparable across traces, so classical amplitude-ratio discriminants survive only as proxies. Despite this limitation, z-scoring removes non-physical scale variation,

suppresses amplitude difference from station gain and is common in this domain.[39, 40]

B. Models and Ensemble Design

1. Model zoo

Five classifiers span the architectural space of the reviewed literature (Tab. I). Inductive bias diversity is the design intent amongst this set: models consuming raw waveforms, spectrograms and hand-crafted features respectively should fail differently, enabling useful ensembling and informative cross-member attribution comparison (**Q4**).[41, 42] A sixth, **SWP-Net**, is original to this work and is inspired by the spectrogram, waveform and physics branches of the two strongest reviewed designs.

2. Geometric constraints

Source–receiver geometry (distance and depth) is deliberately withheld from every model. A deep event recorded nearby and a shallow event recorded far away both present a longer, more attenuated path, therefore we consider these parameters poorly constrained. Depth was trialled as a meta-learner input and the resulting analysis is reported in Sec. III C (**Q2**).[43]

3. Calibration, and stacking

Differently-trained architectures do not necessarily emit probabilities of comparable confidence.[44] Member outputs are therefore temperature-calibrated on validation before combination to adjust for confidence. Temperature scaling fits a single scalar T per member, minimising validation negative log-likelihood, and rescales that member’s log probabilities by T before re-normalising and stacking with the meta-learner.[45]

The meta-learner consumes the 36 calibrated class probabilities, uncertainty features (inter-member disagreement, mean predictive entropy, per-member maximum confidence – App. K), and a wave-feature block to contextualise probabilities to a specific waveform.[46, 47] This lets the meta-learner learn which member to trust and when, allowing it to outperform the calibrated members (see Sec. III A). The wave-feature block admits hand-crafted physics descriptors, frozen self-supervised embeddings, or both.

C. Experimental design

1. Spatial generalisation

If a classifier has learned source physics, the region in which it was trained should not determine where it can

be applied.[26] This is the most direct test of **Q3** we can perform, because in any real corpus, region and archive are potential class confounds. Here we cut the world into three disjoint longitudinal sectors (Fig. 1), each receiving its own split (audited for cross-sector leakage) using the original data-splitter, so the forward-in-time protocol holds. A model trained on each sector is evaluated on every sector’s test split, giving a 3×3 transfer matrix whose diagonal is in-sector performance.

Macro-F1 averages only over classes present in the labels, so a sector holding two easy classes would outscore one holding five. An evaluable class set is therefore fixed globally as those with at least 30 traces in every sector’s train and test split, yielding **nuclear_explosion**, **earthquake** and **noise**. Every number here is a three-class macro-F1 and is not comparable to Sec. III A.

2. P-pick sensitivity

Uncertainty in P-onset is a consistent feature in seismological event discrimination.[48] A model learning source-physics should be agnostic to precisely where the onset appears in the window provided pre- and post-onset discriminatory structure is still preserved (**Q4**). In this experiment (Sec. III F), all models are frozen and only the test data is re-windowed. For each trace the pick is shifted by a fraction of that trace’s own estimated P-wave duration – the time for the trace’s smoothed Hilbert-envelope amplitude to decay from its post-pick peak to halfway back to the pre-pick noise floor (full method in App. L) – and the 60s window is re-cut from the raw waveform and re-normalised.[49] We consider 3,346 stored-pick test traces used in this experiment (800 each bar **volcanic_eruption** which used 146).

3. Training volume sensitivity

Training data volume is cut to investigate the effect of trace scarcity on performance.[50, 51] High-value classes are often rare and therefore we stress-test **Q1**, addressing whether the ensemble’s performance is a property of data abundance. Training rows are subsampled to fractions of $\{1.0, 0.5, 0.25, 0.1\}$, where for each fraction the same seed is used to conduct a uniform random draw of traces without replacement from the full training pool. This procedure operates at trace level. Validation and test sets are all left untouched, so any measured drop is attributable to that class’s training volume alone. All six members are retrained under each condition and combined into an ensemble (Sec. II B 3). One event can be recorded at many stations; this means a trace-level cut is not necessarily commensurate with an event-level cut (App. N).

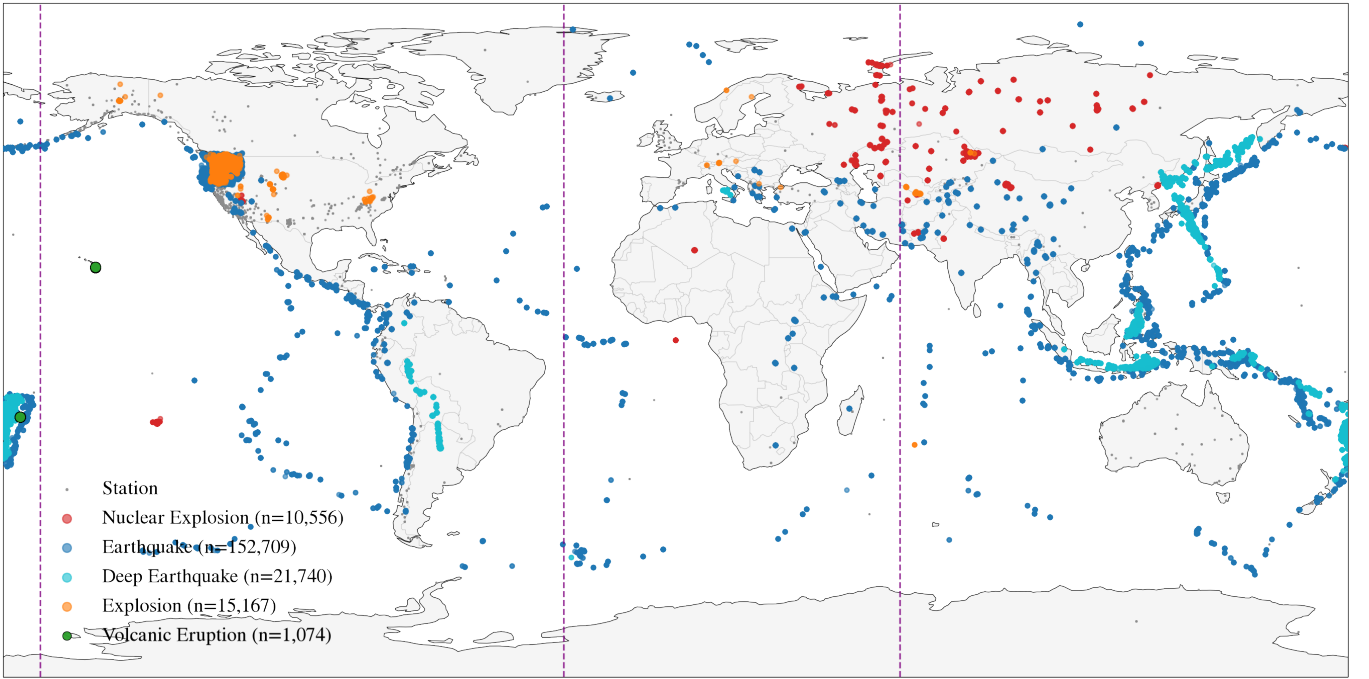


FIG. 1: Global event locations in the processed dataset (436,673 traces total of which 235,426 are **noise**). Stations are shown in grey. The three longitude sectors used for spatial generalisation experiments are marked by purple dashed lines (denoted: **Americas** (the sector containing the legend), **EMEA**, and **AsiaPac** from left to right.). Volcanic eruption markers are enlarged for visibility – these correspond to two systems Hawaii and Tonga.

4. Self-supervised representations

Self-supervised features are the frozen bottleneck activations of two masked autoencoders, one trained on waveforms and one on spectrograms. If unlabelled structure separates the abundant classes, there is no need to process them under the ensemble (**Q1**). [52] Self-learned embeddings therefore can serve two separate roles: meta-learner features [53] and cascading functionality for pre-screening to short-circuit traces out of high data-volume classes prior to reaching the main ensemble.

Two masked autoencoders (MAEs) are pretrained without labels, one per representation. A 2-D convolutional encoder-decoder reconstructs randomly masked patches of the log-spectrogram (8x8-pixel patches, 50% masked per trace), a 1-D convolutional counterpart does the same for randomly masked time-chunks of the raw waveform (1s chunks, 40% masked per trace). Each produces a frozen 128 dimensional embedding per trace.

The cascade is evaluated on 20,000 test traces sampled uniformly from the test split (per-class breakdown in App. F). Each stage targets one abundant class (**noise** or **earthquake**) and decides whether an incoming trace belongs to it, short-circuiting it away from the full ensemble. The short-circuit threshold is fixed on validation to sit below the lowest score (most noise or earthquake-like other class – we adopt a 10% margin) achieved by any trace from the guard classes (the five others) so that no guard-class trace slips through. Three scorers are com-

pared: a Mahalanobis distance (Euclidean distance to the class mean, scaled by that class’s covariance) [54] in one class’s frozen embedding space, and two single-class generative models (VAE and diffusion) judged on reconstruction error.

III. RESULTS AND DISCUSSION

A. In-distribution performance

The four deep models cluster at macro-F1 0.79–0.88 against 0.58–0.61 for the two classical members. All six are strong on the abundant classes – **noise** is completely resolved by the deep models (F1 > 0.99) and almost completely by the classical models (F1 > 0.96). Deep models therefore extract information the hand-crafted feature set discards – reinforcing our question about what a model “sees” (**Q4**).

The deployed meta-learner (stacked ensemble) combines hand-crafted (App. D), self-supervised (Sec. II C 4, App. J), calibrated member probability and uncertainty features attaining macro-F1 0.9104 against 0.8757 for the best single member: **explosion** rises from 0.7643 to 0.7988 and **deep_earthquake** from 0.9087 to 0.9482. Event-level aggregation (averaging per-trace probabilities within each event and taking the argmax) excludes any event with a single contributing trace (**noise** is always singleton by construction). This retains 6,257 of

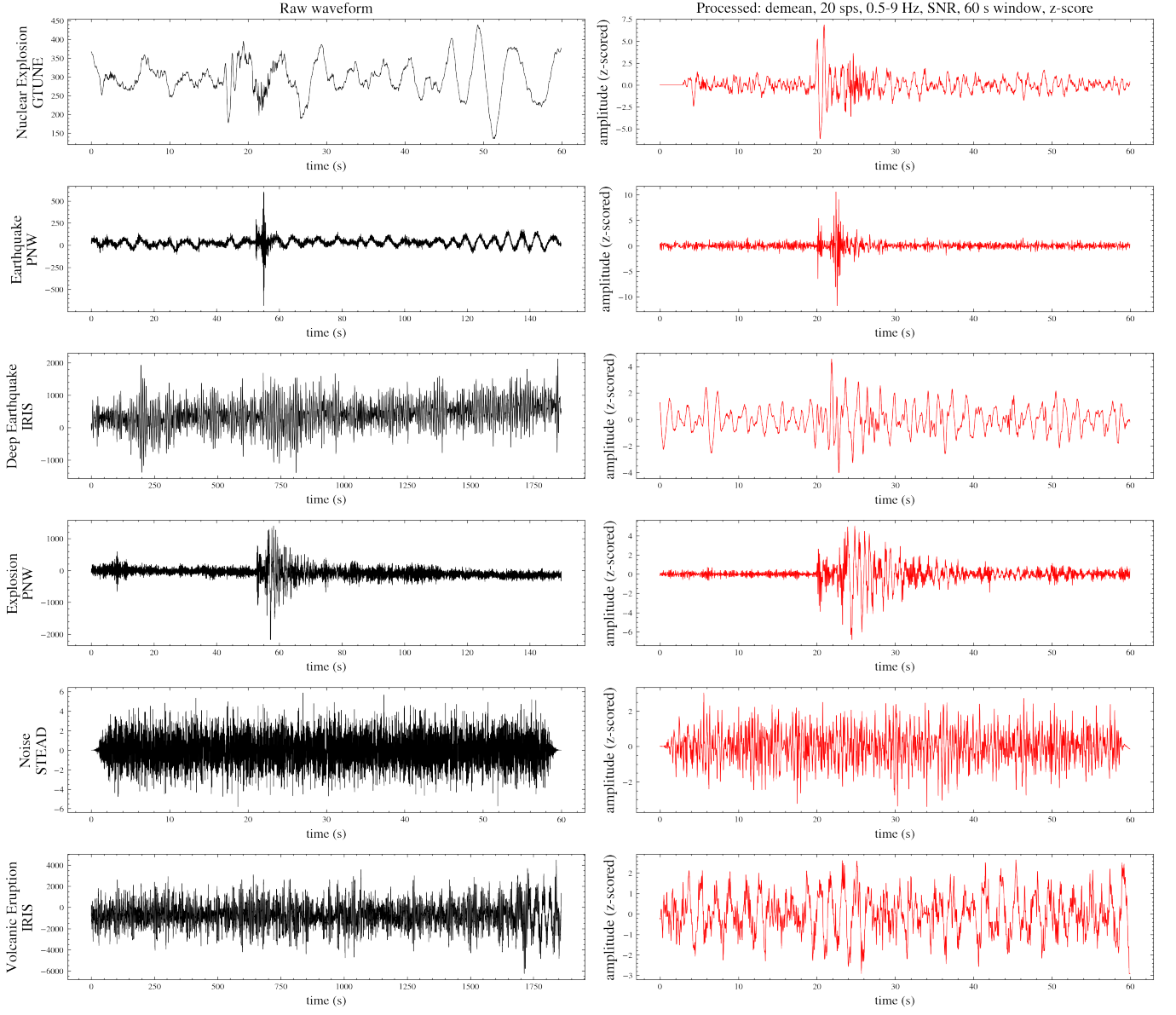


FIG. 2: Raw waveforms adjacent to the processed trace passed to the models for all 6 classes. In order: **nuclear_explosion** (GTUNE), **earthquake** (PNW), **deep_earthquake** (IRIS), **explosion** (IRIS), **noise** (STEAD), **volcanic_eruption** (IRIS).

42,728 events (App. H, Tab. XIII). Inter-trace corroboration (Tab. II, Fig. 3) shows near complete resolution of our priority class (**nuclear_explosion**) is achieved when an event is characterised by $n_{\text{traces}} > 1$ ($F1_{\text{Nuc}} > 0.98$).

Confusion matrices (Fig. 3) show strong recall across all five classes retained under the $n_{\text{traces}} > 1$ restriction. Precision is weakest on **explosion** (0.82, $n_{\text{event}}=676$) and **volcanic_eruption** (0.80, $n_{\text{event}}=4$, small n_{event} imposes skepticism on stability for both precision and recall) – App. H, Tab. IX and Tab. XIII for precise trace counts. **nuclear_explosion** appears to conflate only with **explosion** (0.01) under recall and **earthquake** and **explosion** (0.02 and 0.01), under precision; no other

classes touch **nuclear_explosion** making these the operationally significant hurdles (**Q1**).

B. Model sensitivity

We next diagnose brittle model configurations across eighteen sweeps of all six members (68 grid-points). Every sweep’s macro-F1 stays within 0.0089–0.1161 except one. The CNN’s dropout spread reaches 0.7471, collapsing from 0.8385 at dropout 0.0 to 0.0914 at 0.5. **Phys-CNN** and **ViT** are most stable never exceeding 0.0250 across an order of magnitude in learning rate and their respective

TABLE I: The six ensemble members. Physics denotes four scalar features (`hf_lf_energy_ratio`, `dominant_frequency`, `spectral_centroid`, `spectral_flux`), standardised on train only. “features” denotes a hybrid set of physical and statistical features (see App. D). **SWP-Net** is original to this project, the remaining five are reimplementations of the reviewed literature under a common harness and pipeline, which is what makes them comparable. Tuned configurations are in the App. G, Tab. XI. CNN: Convolutional Neural Network; Phys-CNN: Physics informed CNN; ViT: Vision Transformer; SVM: Support Vector Machine; XGBoost: Extreme Gradient Boosting; SWP-Net: Spectrogram, Waveform, Physics Network

Model	Paper	Input	Architecture
CNN	[19]	waveform	1-D CNN, 9 blocks of Conv-BatchNorm-ReLU-Dropout-MaxPool, linear head
Phys-CNN	[21]	waveform + physics	1-D CNN waveform branch and a scalar physics branch, embeddings concatenated and passed to a shared decision head for linear projection to classes
ViT	[20]	spectrogram	Vision transformer on linear-frequency log-amplitude spectrogram patches
SVM	[23]	features	RBF-kernel SVM, calibrated, capped at 50,000 training samples
XGBoost	[22]	features	XGBoost on hand-crafted spectral and temporal features
SWP-Net	original	spec. + wfm. + physics	Spectrogram branch (2-D CNN, pooled over time, frequency bins attend to one another via self-attention), waveform CNN branch, and scalar physics features, all concatenated into a shared decision head

TABLE II: Test-split performance, per member and for the stacked ensemble. Member values are as-trained (uncalibrated); the ensemble consumes temperature-calibrated probabilities alongside hand-crafted and self-learned features. Temperature scaling rescales every class’s score by the same factor so it cannot change which class a member predicts – each member’s calibrated accuracy and F1 is therefore identical to its uncalibrated output. The direct comparison between ensemble and individual member is therefore 0.9104 against **SWP-Net**’s 0.8757. The best per column is emboldened and the best individual member is underlined. F1 on **noise** is omitted due to near-complete resolution. Meta-learner outputs are displayed at the trace and event-level (event-level separated due to $n_{\text{traces}} > 1$). Individual model outcomes are reported at the trace-level. NLL is negative log-likelihood. We also report Platt-calibrated outcomes: App. H. Values are reported to four decimal places to support delta comparisons, seed refit variability, and allow approximate standard-error assessment App. M.

Model	Acc. $\uparrow$	Macro-F1 $\uparrow$	Bal. Acc. $\uparrow$	NLL $\downarrow$	F1 _{nuc} $\uparrow$	F1 _{eq} $\uparrow$	F1 _{deep} $\uparrow$	F1 _{exp} $\uparrow$	F1 _{volc} $\uparrow$
SWP-Net	<u>0.9667</u>	<u>0.8757</u>	0.8705	0.1872	<u>0.8962</u>	<u>0.9684</u>	<u>0.9087</u>	<u>0.7643</u>	<u>0.7212</u>
Phys-CNN	0.9640	0.8620	<u>0.8739</u>	0.1245	0.8867	0.9683	0.8883	0.7439	0.6894
ViT	0.9614	0.8412	0.8340	0.2226	0.8532	0.9657	0.8921	0.7483	0.5929
CNN	0.9364	0.7872	0.8656	0.2232	0.7617	0.9453	0.8188	0.6292	0.5749
XGBoost	0.8834	0.6108	0.6867	0.3780	0.5196	0.9229	0.6497	0.4734	0.1345
SVM	0.8905	0.5796	0.5934	0.4728	0.4896	0.9233	0.6474	0.4525	0.0000
Ensemble – Trace	0.9770	0.9104	0.9437	0.1530	0.9542	0.9771	0.9482	0.7988	0.7866
Ensemble									
Event-level ($n_{\text{trace}} > 1$)	0.9717	0.9451	0.9817	0.1160	0.9847	0.9836	0.9869	0.8815	0.8889

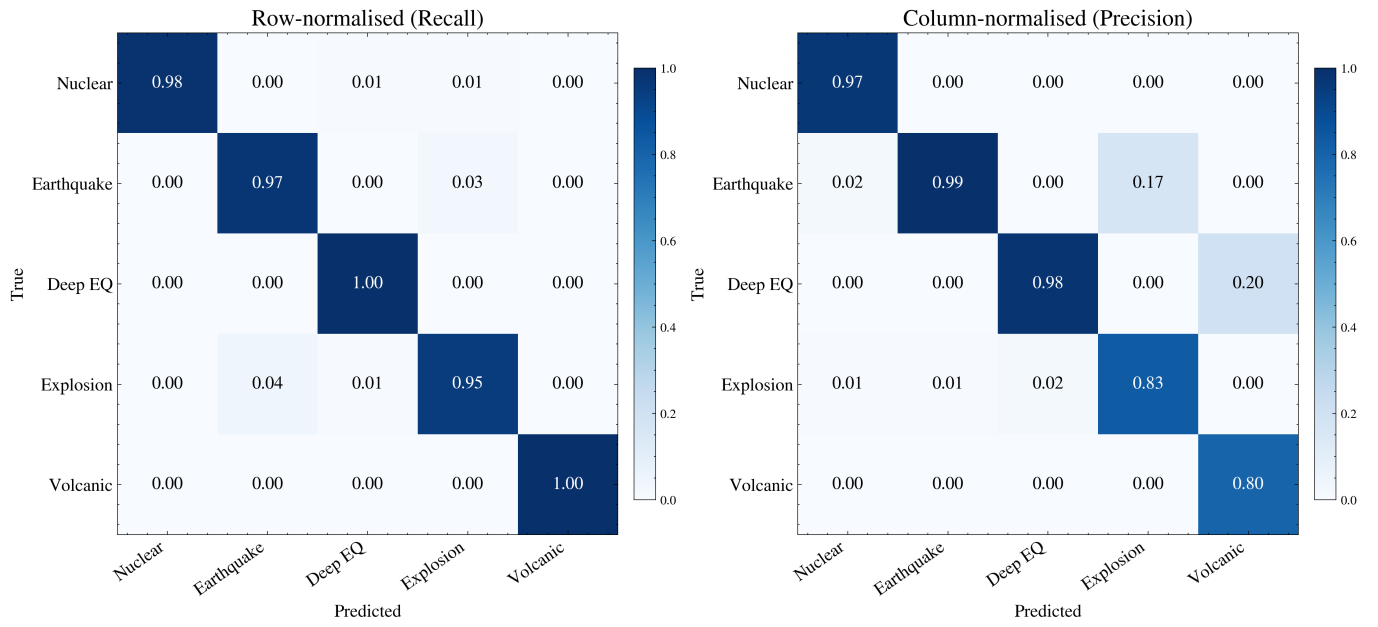


FIG. 3: Confusion matrices denoting recall (left) and precision (right) of the event-level ($n_{\text{trace}} > 1$) stacked ensemble (meta-learner) across 6,257 events (For trace counts: App. XIII).

drouput sweeps (0.1–0.7 and 0.0–0.3) – with one exception.

Swept over learning rates, CNN moves 0.0595 across a 30-fold range, and its weight-decay moves 0.0574 across three orders of magnitude. The classical members are mildly more sensitive than the deep ones (excluding CNN dropout): classical spreads range 0.0650–0.1161 against 0.0089–0.0624 for the deep members. Full grids are in App. B. Hyperparameter choice is therefore, not likely to be a confound for **Q1**.

C. Depth as a leakage diagnostic

Q2 asks whether the system takes a metadata shortcut when one is offered, and whether that can be detected before deployment. To investigate this, four ensemble (trace-level) seed variations were fitted differing only in the depth feature handed to the meta-learner: none (A, the deployed configuration), a three-way shallow (< 10 km)/deep (≥ 10 km)/unknown bin (B); two further continuous-depth and reassignment variants are in App. O. Adding depth improves trace macro-F1 substantially: 0.8774 (A) to 0.9428 (B) with nuclear F1 rising from 0.9374 to 0.9829. Across the seeds, A and B do not overlap therefore the effect is not noise.

Metadata audits indicate that gains arise from an archival shortcut as opposed to additional seismological information. First, **deep_earthquake** was harvested by a depth query disjoint from **earthquake**’s – measured maximum 102.2 km against measured minimum 300.0 km – so a single threshold separates the two. Second, **noise** and **nuclear_explosion** come from archives carrying no catalogue depth, so depth by imputation and convention is fixed to 0 km for both. Since the members almost completely resolve noise ($F1 > 0.99$), both conditions needs only learn “depth zero and not noise, therefore nuclear” – inflating output through metadata provenance rather than waveform character.

We report this as a leakage diagnostic rather than a gain, and retain condition A. This experiment demonstrates concretely that catalogue metadata can encode both label definitions and source-archive identity, and retrospectively justifies geometry exclusion. It is the best illustration of one of this report’s central concerns: a feature that improves headline metrics while teaching the system nothing about source-physics. Here we show a shortcut is taken and ablation exposed it, answering both parts of **Q2**.

D. Spatial generalisation

Table III reports model spatial transfer performance (**Q3**). The sectors are severely imbalanced: the **Americas** hold 82% of traces. The **Americas** model saw 240,420 training traces and still degrades off-sector; Sec. III G shows training volume is not the cause. The

TABLE III: Spatial transfer, **three-class macro-F1** over the evaluable set (**nuclear_explosion**, **earthquake**, **noise**); not comparable to Table II. Upper block: mean in-sector against mean cross-sector performance per model. Lower block: holding the test sector fixed to control for test-set difficulty, averaged over the six individual members. Full 3×3 matrices and per-sector class priors are in Appendix C

Model	In-sector	Cross-sector	Gap
Phys-CNN	0.9165	0.6773	0.2392
SWP-Net	0.9094	0.7087	0.2007
XGBoost	0.7873	0.5799	0.2073
CNN	0.9065	0.6771	0.2295
SVM	0.8092	0.5930	0.2162
ViT	0.9210	0.6867	0.2343
Fixed test sector			
Americas	0.911	0.352	0.559
EMEA	0.861	0.834	0.027
AsiaPac	0.853	0.775	0.078

collapse of the **Americas** as a fixed test set can be explained by archival homogeneity: 97.7% of **earthquake** traces in the **Americas** are from PNW, whereas **EMEA** and **AsiaPac earthquake** are 100% IRIS. It is **earthquake** that drives this collapse ($F1_{eq}$ of 0.0135 and 0.0520 for training on **EMEA** or **AsiaPac**, and tested on **Americas**), delineating that archive identity becomes an exploitable shortcut if training data is drawn from a single source.[55] Considering **Q3**, the training region does constrain model application, but the cause is archival and diversification can mitigate this.

E. Forward-in-time generalisation: North Korea

A primary objective of this work was establishing a forward-in-time prediction protocol capable of geographic generalisation. Separate from the spatial generalisation experiment performed above, here the ensemble is trained on the full corpus and tested on **nuclear_explosions** from *Punggye-ri* – North Korea, 2006–2017. This offers the cleanest examination of our priority class forward in time and across an unseen region: 1,363 traces across five (out of six[56]) catalogued nuclear tests (withheld from train and val) reveal a 92.5% Platt calibrated recall and a 89.8% temperature calibrated recall at the trace-level using the stacked meta-learner. A separate analysis of model attribution can be found in App. A. This result clearly demonstrates the ensemble’s operational value (**Q1**), and provides strong positive evidence for **Q3**.

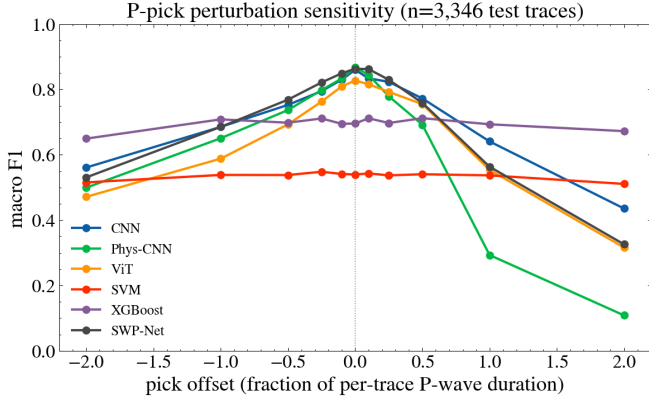


FIG. 4: P-pick sensitivity. Each model is frozen and only the test data is re-windowed. The x-axis is the pick offset as a fraction of the estimated P-wave duration, and the y-axis is macro-F1. Every deep model peaks at zero offset and falls away either side, while the classical members are flat. We take 800 samples for each class bar `volcanic_eruption` which only has 146 traces in the test set (total 3,346).

F. P-pick sensitivity

Fig. 4 shows a clean architecture separation between the two model families. Every deep model (not tuned) peaks at zero offset – Phys-CNN 0.8682, SWP-Net 0.8633, CNN 0.8600, ViT 0.8272, – and falls away either side. A negative shift may maintain the impulsive onset in the post-pick 40s where forward shift has only a 20s pre-pick window to maintain the discriminatory signature. At a two-duration offset Phys-CNN reaches 0.1084, below the ~ 0.2 that balanced random guessing over five classes would give. Therefore misalignment actively misleads the model. Both classical members are flat: XGBoost moves 0.6723 to 0.6494 and SVM 0.5155 to 0.5108 across the same range. This is a partial failure under **Q4**. Window position is only centering – it is never handed to the models – so a model reading source structure should be agnostic to this (assuming structure survives re-cutting).

G. Data-volume sensitivity

Under aggressive (90%) trace-volume cutting, members stay flat on the abundant classes (test F1 within 0.0109 for five of six models). The SVM loses 0.1478 on `earthquake` and collapses from 0.4525 to 0.0047 on `explosion`; CNN loses 0.2293 on `nuclear_explosion`. The ensemble is able to offset this: at 10% volume the best surviving member on `nuclear_explosion` (Phys-CNN, 0.8500) sits below the ensemble (0.9250), and on `explosion` the ensemble moves only 0.7381 to 0.7008. The meta-learner reweights toward whichever member remains informative rather than inheriting its weakest member’s failure: under **Q1** the ensemble’s advantage is

a property of member diversity, not of data abundance. Because the cut is uniform at trace level it measures dependence on repeated observation of an event rather than on event diversity. Event volume reduction remains high value future work (App. N).

H. Cascading: self-supervised representations

The cascade tests the throughput premise of Sec. I – whether unlabelled structure can triage abundant classes without leaking priority classes. For the `noise` stage, the diffusion model performs below change against all five guard classes (AUC 0.05–0.26), rating `deep_earthquake` as more noise-like than `noise` itself; the Mahalanobis scorer is the only one to serve `noise` well, screening 23.7% of traces at zero leakage.

For the `earthquake` step, the diffusion model happens to be the best performer: screening 36.0% of traces at 0.40% leakage against Mahalanobis’ 23.5% at 0.13% and the VAE’s 16.7% at 5.2% (guard floors at the 2nd, 1st and 5th percentile of guard-class validation scores respectively). Despite non-trivial triage at low leakage we do not recommend deployment: 2 of the 50 leaked traces were `nuclear_explosion`, and a demonstrated loss of the priority class prohibits this screening step.

I. Model explainability

Attribution methods for XAI are dispatched by model family and provide an interpretable assessment of what drives model decisions (App. E) – the direct test of **Q4**. Here we consider a nuclear case study displaying a confident success where all members get it right, and a dangerous failure where all 6 members miss the `nuclear_explosion`. Attribution methods are run on all six members.

The confident-success trace is teleseismic (16,250 km) and stems from a 1995 French test (*Ploutos*, yield ~ 110 kt). The failure trace is regional (1,001 km) and comes from a 1989 UK/US test (Nevada Test Site, *Barnwell*, yield 120 kt).[57, 58] Its catalogued P-pick is at 20 s and corresponds to a modest waveform onset. The trace’s most prominent feature is a sharp peak at ~ 30 s.

Saliency[59] and integrated gradients[60] disagree throughout, on both traces. Vanilla saliency is vulnerable to the local loss surface: a region the model is already confident about has locally saturated gradients. Integrated gradients accumulate gradients along a path from a baseline and are more robust to this.[61]

On the success case, attribution generally aligns with physically plausible structure (Fig. 5, Fig. 6). The ViT’s attention rollout[62] shows a full-spectrum band at the 20 s onset, concentrated at high frequency, with dispersed mild attention later. SWP-Net’s Grad-CAM[63] shows a strong nucleated attribution just before the onset (10–20 s, 0–4 Hz). High frequency attribution presents mid-

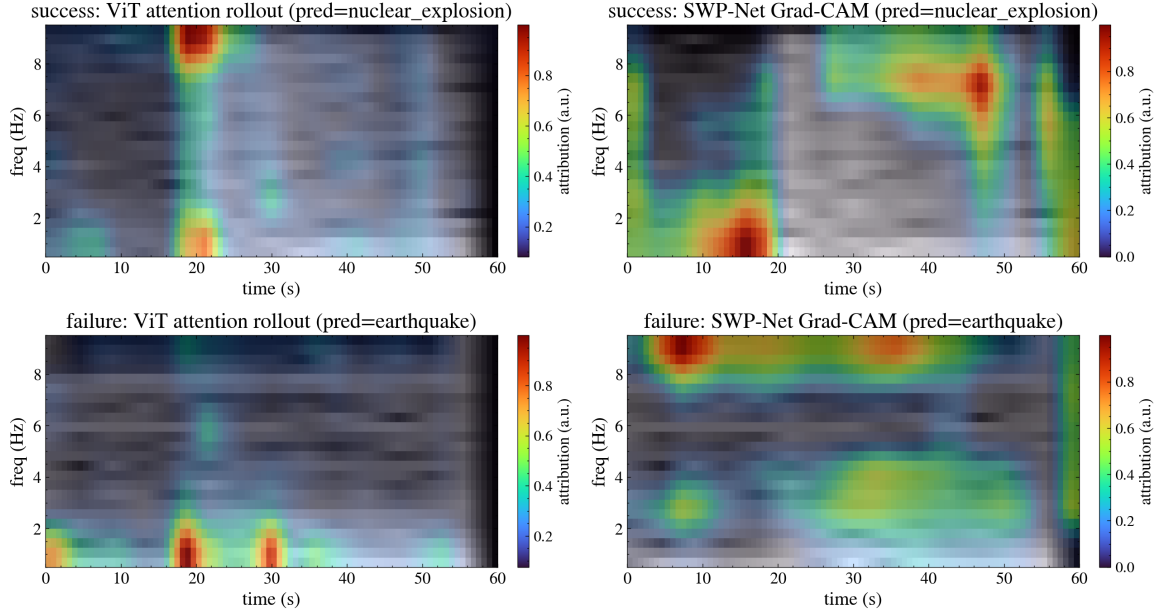


FIG. 5: Spectrogram attribution for the confident-success (top) and confident-failure (bottom) traces: ViT attention rollout (left) and SWP-Net Grad-CAM (right).

to-late time (25–50 s) which expands into a vertically extensive band, peaking between 6–8 Hz. This indicates attention to the coda and might corroborate the pre-onset attribution; the model may be observing a quiet build-up followed by a large range of frequency decay – an example of differing inductive bias.

The waveform branches (Fig. 6) behave similarly: saliency mass concentrated in the quiet pre-onset region, with integrated gradients and time occlusion tracking the waveform envelope well; SWP-Net’s time occlusion specifically peaks late (40–60 s), consistent with coda importance. The Phys-CNN SHAP branch (Fig. 7) shows no feature attribution because the prediction is completely resolved by the waveform itself. Overall, these attribution methods agree that the onset is the most important feature but attention to coda also matters – a physically plausible conclusion.

In the failure case, model convergence is weaker. The ViT’s attention rollout (**earthquake** prediction) shows a dispersed band at the 20 s onset, and a second strong low-frequency nucleation at ~30 s aligning with the waveform’s largest peak – plausibly mimicking P then S-wave attention, and a potential explanation for the **earthquake** call. SWP-Net’s Grad-CAM disagrees: a strong high-frequency nucleation (8–9.5 Hz, 5–10 s) extends with reduced intensity out to ~48 s, alongside a weaker low-mid frequency (2–4 Hz) region early and a diffuse mid-frequency band from 20–55 s.

The waveform branches show some agreement: CNN’s integrated gradients display early-time attention alongside its saliency, and its time occlusion responds most to the ~30 s peak. SWP-Net’s waveform branch shows the same early saliency mass and better envelope-tracking in-

tegrated gradients, but its time occlusion instead peaks late (~46 s), suggesting coda rather than onset sensitivity. Phys-CNN’s waveform branch shows the lowest saliency mass and integrated gradients that again track the envelope, but time occlusion is elevated across the whole window rather than one clear peak. Hand-crafted and physics-branch SHAP disagree on which feature leads (Fig. 7). The XGBoost shows modest attribution overall, but the SVM and Phys-CNN SHAP both show dominant features. Despite some disagreement, the waveform branches still concentrate attribution on the 20 s onset and the ~30 s peak.

Meta-learner attribution changes significantly between success and failure modes. Member probability outputs dominate the success trace (~45%) but collapse to ~1% on the failure trace, where the SSL embeddings absorb the rest (hand-crafted and uncertainty features show minor attribution). This suggests that discarding the member outputs may be damaging, and that although SSL features can capture discriminative structure, they also might actively mislead the meta-learner away from members consensus.

Across both success and failure, models are plausibly attending meaningful structure (Q4), but this alone proves insufficient. On the success spectrogram attention rollout and Grad-CAM attend the same ~20 s onset, while on the failure spectrogram both arguably attend physical features but still fail. Although all six members unanimously vote **earthquake** on the failure mode the meta-learner discards the members’ unanimous, albeit wrong, consensus in favour of SSL features that here mislead rather than help. Different architectures are demonstrably not making the same mistake for the same reason,

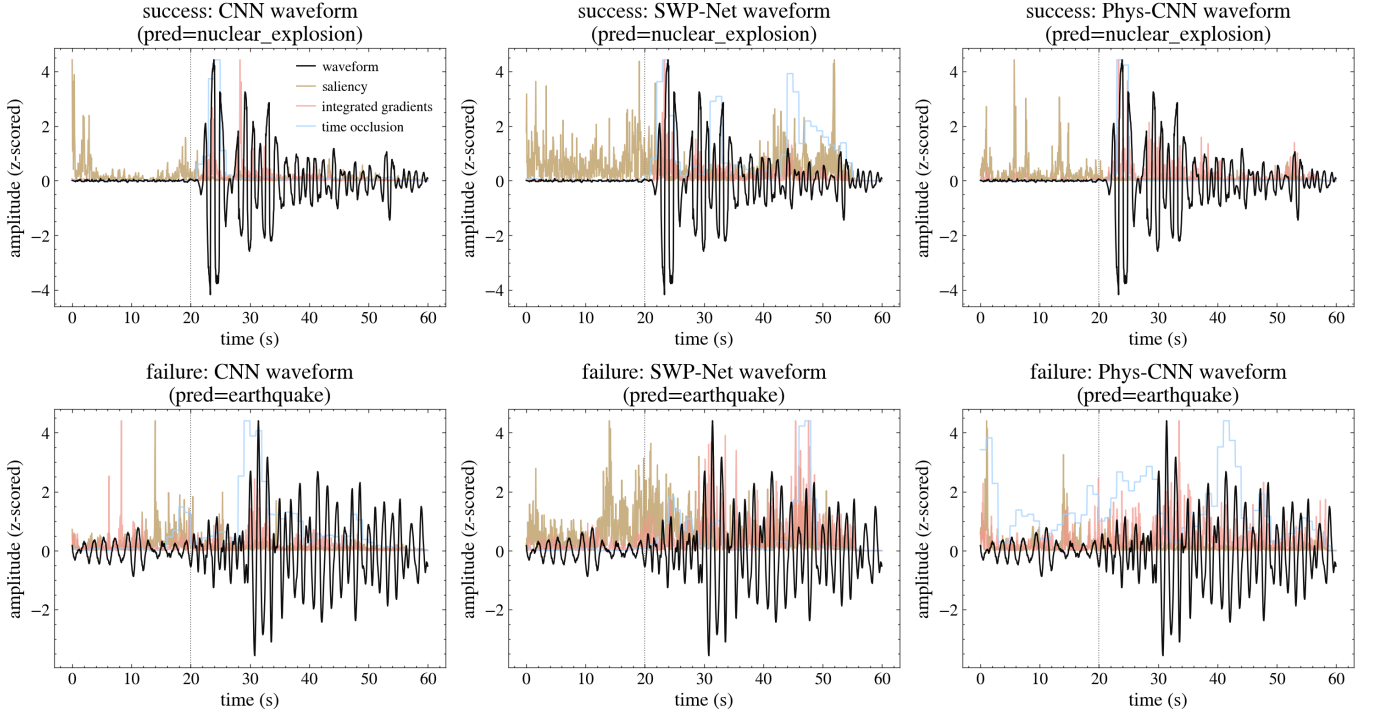


FIG. 6: Waveform-branch attribution (saliency, integrated gradients, time occlusion) for CNN, SWP-Net, and Phys-CNN on the success (top) and failure (bottom) traces.

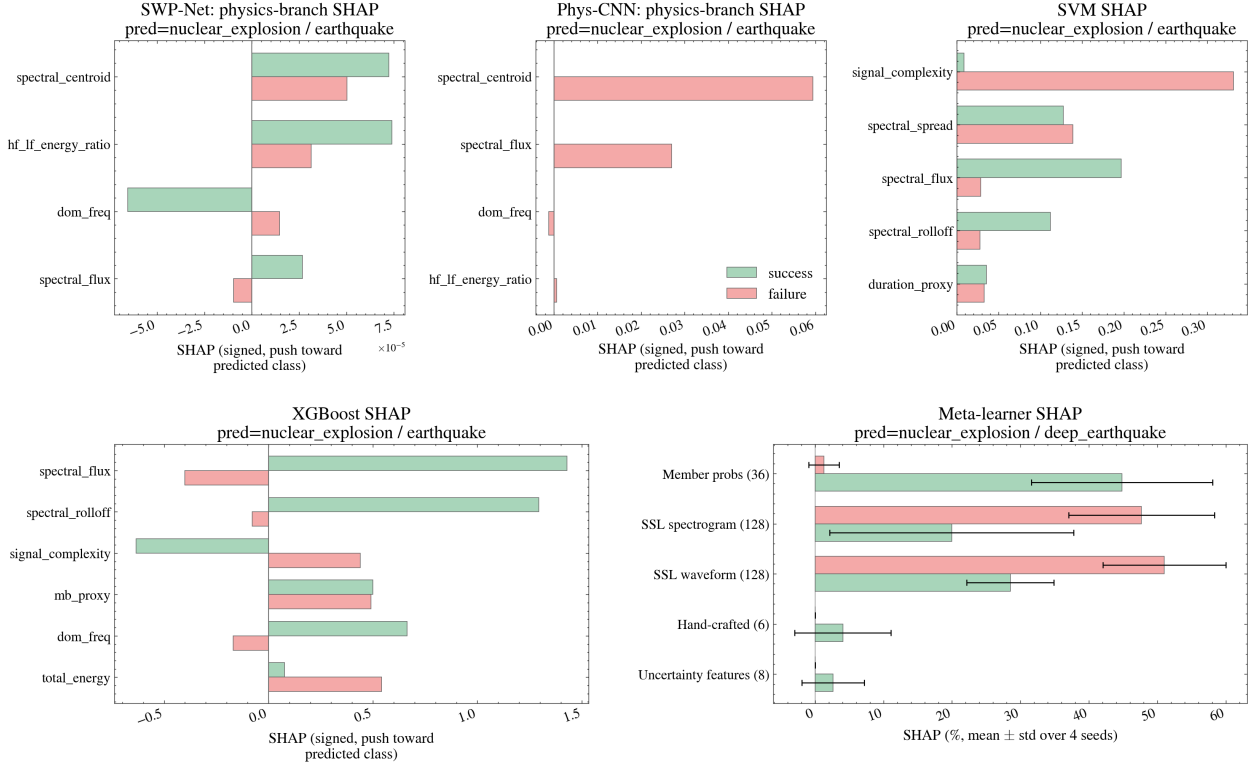


FIG. 7: SHAP attribution on the confident-success (green) and confident-failure (red) traces, paired per feature: SWP-Net and Phys-CNN physics-branch SHAP, XGBoost and SVM SHAP over the top contributing features from either trace, and the meta-learner’s own grouped-feature SHAP (mean $\pm$ std over 4 seeds error bars).

consistent with inductive-bias diversity.

IV. METHODOLOGICAL LIMITATIONS

The most consequential limitation of our work is the omission of horizontal channels: P/S amplitude ratios,

which require three-component data and are not common across the five archives. Mine collapses are also excluded due to data sparsity despite being a critical CTBT class – their implosional source producing waveforms analogous to nuclear explosions.[1]

We also admit procedural limitations. Per-trace z-scoring removes absolute amplitude. Platt calibration and the meta-learner are fit on the same validation split sequentially. Most experiments are single-seed: only depth-binning was swept across 4 seeds. The presented nuclear case-study is illustrative – not an exhaustive population-level claim (Q4). Of the four main questions, Q3 was hardest to answer. The sector protocol cannot preserve all six classes, so spatial transfer is established only over the three evaluable ones. Data-volume sensitivity is measured by uniform random subsampling at the trace level; event-level subsampling would better identify dependence on event diversity. High quality nuclear explosion traces are rare; we therefore use only one aggregated archive (GTUNE), and are not able to fully verify the absence of an archival shortcut for our priority class.

V. CONCLUSION

Deep and classical models could become viable tools in CTBT verification when operated as a human-in-the-loop workflow. We have introduced a six-member stacked ensemble for global multi-class seismic event classification (trace-level: macro-F1 = 0.9104, $F1_{\text{Nuc}}$ = 0.95; event-level: macro-F1 = 0.95 $F1_{\text{Nuc}}$ = 0.98) under a forward-in-time prediction regime (Q1). To achieve this we unified 5 seismic archives and combine a diverse model set – motivated by inductive bias – from prior literature with a novel architectural contribution. We demonstrate ensemble robustness with general resistance to data-volume drops under uniform random sub-sampling of the training pool, and insensitivity to a range of hyperparameter choices (Q1). We display the potential for self-learned features to be useful for high-volume class triage (Q1). We justify the exclusion of geometric features from the ensemble via depth-leakage experimentation (Q2). Spatial generalisation is shown to be possible if the data that forms the training corpus is archivally heterogeneous (Q3). A case study on the North Korean nuclear tests from 2006–2017, show the ensemble is capable of forward in time generalisation to an unseen region (Q1, Q3). An explainability suite diagnoses a nuclear case-study for success and failure modes across the model ensemble (Q4). Returning to our central question of whether a discriminator learns source-physics or spurious archive, path and site effects: we demonstrate that models are capable of learning source-physics, even if they do not converge on the same evidence, therefore, we accept our initial hypothesis. High-value future work includes repeating the volume-sensitivity experiment at the event level rather than the trace level, to isolate dependence on event diversity from repeated observation of the same

event, extending the explainability suite to a statistical population-level study to definitively locate where attribution mass resides, and sourcing `nuclear_explosion` data from additional independent archives.

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- [1] D. Bowers and N. D. Selby, Forensic seismology and the comprehensive nuclear-test-ban treaty, *Annual Review of Earth and Planetary Sciences* **37**, 209 (2009).
- [2] S. R. Taylor, M. D. Denny, E. S. Vergino, and R. E. Glaser, Regional discrimination between nts explosions and western u.s. earthquakes, *Bulletin of the Seismological Society of America* **79**, 1142 (1989), <https://pubs.geoscienceworld.org/ssa/bssa/article-pdf/79/4/1142/5335030/bssa0790041142.pdf>.
- [3] P. D. Marshall and P. W. Basham, Discrimination between earthquakes and underground explosions employing an improved ms scale, *Geophysical Journal International* **28**, 431 (1972), <https://academic.oup.com/gji/article-pdf/28/5/431/1755175/28-5-431.pdf>.
- [4] J. L. Stevens and S. M. Day, The physical basis of mb: Ms and variable frequency magnitude methods for earthquake/explosion discrimination, *Journal of Geophysical Research: Solid Earth* **90**, 3009 (1985).
- [5] N. D. Selby, P. D. Marshall, and D. Bowers, mb: M s event screening revisited, *Bulletin of the Seismological Society of America* **102**, 88 (2012).
- [6] W. R. Walter, K. M. Mayeda, and H. J. Patton, Phase and spectral ratio discrimination between nts earthquakes and explosions. part i: Empirical observations, *Bulletin of the Seismological Society of America* **85**, 1050 (1995).
- [7] H. J. Patton, Regional magnitude scaling, transportability, and ms: mb discrimination at small magnitudes, pure and applied geophysics **158**, 1951 (2001).
- [8] H. E. Hartse, S. R. Taylor, W. S. Phillips, and G. E. Randall, A preliminary study of regional seismic discrimination in central asia with emphasis on western china, *Bulletin of the Seismological Society of America* **87**, 551 (1997).
- [9] D. Anderson, H. Patton, S. Taylor, J. Bonner, and N. Selby, Sources of error and the statistical formulation of ms: mb seismic event screening analysis, *Pure and applied geophysics* **171**, 537 (2014).
- [10] N. R. Council, Policy, G. Affairs, C. on Reviewing, and U. T. I. R. to the Comprehensive Nuclear Test Ban Treaty, *The comprehensive nuclear test ban treaty: Technical issues for the United States* (National Academies Press, 2012).
- [11] L. R. Sykes, Dealing with decoupled nuclear explosions under a comprehensive test ban treaty, *Monitoring a Comprehensive Test Ban Treaty*, 247 (1996).
- [12] Q. Kong, D. T. Trugman, Z. E. Ross, M. J. Bianco, B. J. Meade, and P. Gerstoft, Machine learning in seismology: Turning data into insights, *Seismological Research Letters* **90**, 3 (2019).
- [13] S. M. Mousavi and G. C. Beroza, Deep-learning seismology, *Science* **377**, eabm4470 (2022).
- [14] S. M. Mousavi and G. C. Beroza, Machine learning in earthquake seismology, *Annual Review of Earth and Planetary Sciences* **51**, 105 (2023).
- [15] K. D. Koper, R. Burlacu, A. D. Armstrong, and R. B. Herrmann, Classifying small earthquakes, explosions and collapses in the western united states using physics-based features and machine learning, *Geophysical Journal International* **239**, 1257 (2024).
- [16] R. Geirhos, J.-H. Jacobsen, C. Michaelis, R. Zemel, W. Brendel, M. Bethge, and F. A. Wichmann, Shortcut learning in deep neural networks, *Nature Machine Intelligence* **2**, 665 (2020).
- [17] M. Buda, A. Maki, and M. A. Mazurowski, A systematic study of the class imbalance problem in convolutional neural networks, *Neural networks* **106**, 249 (2018).
- [18] L. Barama, Z. Peng, A. V. Newman, and J. Williams, GTUNE: An assembled global seismic dataset of underground nuclear test blasts, *Seismological Research Letters* **93**, 3514 (2022).
- [19] L. Barama, J. Williams, A. V. Newman, and Z. Peng, Global nuclear explosion discrimination using a convolutional neural network, *Geophysical Research Letters* **50**, e2022GL101528 (2023).
- [20] V. Kasburg, M. Zehner, M. van Laaten, J. Müller, and N. Kukowski, Explainable seismic event discrimination: Improved explainability with vision transformers, *Journal of Geophysical Research: Machine Learning and Computation* **2**, e2025JH000802 (2025).
- [21] Q. Kong, R. Wang, W. R. Walter, M. Pyle, K. Koper, and B. Schmandt, Combining deep learning with physics based features in explosion-earthquake discrimination, *Geophysical Research Letters* **49**, e2022GL098645 (2022).
- [22] H. Li, X. Li, X. Tan, T. Liu, Y. Zhang, J. Liu, and C. Niu, Classification of small sample nuclear explosion seismic events based on MSSA-XGBoost, *Applied Geophysics* **21**, 108 (2024).
- [23] A. Pignatelli, C. Petrucci, V. Vignoli, F. D’Ajello Caracciolo, and R. Console, Deciphering earth’s tremors: a machine learning approach to distinguish earthquakes from explosions, *Journal of Seismology* **29**, 525 (2025).
- [24] S. Kapoor and A. Narayanan, Leakage and the reproducibility crisis in machine-learning-based science, *Patterns* **4** (2023).
- [25] D. R. Roberts, V. Bahn, S. Ciuti, M. S. Boyce, J. Elith, G. Guillerá-Aroita, S. Hauenstein, J. J. Lahoz-Monfort, B. Schröder, W. Thuiller, *et al.*, Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure, *Ecography* **40**, 913 (2017).
- [26] R. Maguire, B. Schmandt, R. Wang, Q. Kong, and P. Sanchez, Generalization of deep-learning models for classification of local distance earthquakes and explosions across various geologic settings, *Seismological Research Letters* **95**, 2229 (2024).
- [27] S. M. Mousavi, Y. Sheng, W. Zhu, and G. C. Beroza, STanford EArthquake Dataset (STEAD): A global data set of seismic signals for AI, *IEEE Access* **7**, 179881 (2019).
- [28] Y. Ni, A. R. Hutko, F. Skene, M. A. Denolle, S. D. Malone, P. Bodin, J. R. Hartog, and A. K. Wright, Curated Pacific Northwest AI-ready seismic dataset, *Seismica* **2**, 10.26443/seismica.v2i1.368 (2023).
- [29] IRIS Data Management Center, IRIS data services, <https://ds.iris.edu/> (2024), seismological Facilities for the Advancement of Geoscience (SAGE) Award of the National Science Foundation under Cooperative Support Agreement EAR-1851048.
- [30] F. Vernon, ANZA regional network (1982).
- [31] IRIS, USArray transportable array (2003).

- [32] Albuquerque Seismological Laboratory (ASL)/USGS, United states national seismic network (1990).
- [33] Albuquerque Seismological Laboratory (ASL)/USGS, Global seismograph network (GSN - IRIS/USGS) (1988).
- [34] Atomic Weapons Establishment (AWE), Blacknest historic digitised data: Various chemical explosions, <https://bdsweb.blacknest.gov.uk/digitised> (2026), last updated 28 April 2026.
- [35] J. R. Peterson, *Observations and modeling of seismic background noise*, Tech. Rep. (US Geological Survey, 1993).
- [36] F. Ardhuin, E. Stutzmann, M. Schimmel, and A. Mangeney, Ocean wave sources of seismic noise, *Journal of Geophysical Research: Oceans* **116** (2011).
- [37] W. Weckesser, Signal processing with scipy: Linear filters, *Signal processing with SciPy: Linear filters* (2016).
- [38] R. Scandone, K. V. Cashman, and S. D. Malone, Magma supply, magma ascent and the style of volcanic eruptions, *Earth and Planetary Science Letters* **253**, 513 (2007).
- [39] S. M. Mousavi, W. L. Ellsworth, W. Zhu, L. Y. Chuang, and G. C. Beroza, Earthquake transformer—an attentive deep-learning model for simultaneous earthquake detection and phase picking, *Nature communications* **11**, 3952 (2020).
- [40] W. Zhu and G. C. Beroza, Phasenet: a deep-neural-network-based seismic arrival-time picking method, *Geophysical Journal International* **216**, 261 (2019).
- [41] T. G. Dietterich, Ensemble methods in machine learning, in *International workshop on multiple classifier systems* (Springer, 2000) pp. 1–15.
- [42] L. K. Hansen and P. Salamon, Neural network ensembles, *IEEE transactions on pattern analysis and machine intelligence* **12**, 993 (1990).
- [43] J. Havskov and L. Ottemoller, *Routine data processing in earthquake seismology: with sample data, exercises and software* (Springer Science & Business Media, 2010).
- [44] A. Niculescu-Mizil and R. Caruana, Predicting good probabilities with supervised learning, in *Proceedings of the 22nd international conference on Machine learning* (2005) pp. 625–632.
- [45] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, On calibration of modern neural networks, in *Proceedings of the 34th International Conference on Machine Learning*, *Proceedings of Machine Learning Research*, Vol. 70, edited by D. Precup and Y. W. Teh (PMLR, 2017) pp. 1321–1330.
- [46] D. H. Wolpert, Stacked generalization, *Neural networks* **5**, 241 (1992).
- [47] L. Breiman, Stacked regressions, *Machine learning* **24**, 49 (1996).
- [48] J. Münchmeyer, J. Woollam, A. Rietbrock, F. Tilmann, D. Lange, T. Bornstein, T. Diehl, C. Giunchi, F. Haslinger, D. Jozinović, *et al.*, Which picker fits my data? a quantitative evaluation of deep learning based seismic pickers, *Journal of Geophysical Research: Solid Earth* **127**, e2021JB023499 (2022).
- [49] R. K. Lobo, Y. H. L. Gaol, D. Y. Fatimah, A. Abdullah, D. A. Zaky, S. K. Suhardja, A. D. Nugraha, Z. Zulfakrizza, S. Widiyantoro, and M. Ramdhan, An attempt to pick teleseismic P wave arrival using envelope and artificial neural network algorithm, *IOP Conference Series: Earth and Environmental Science* **873**, 012059 (2021), the 3rd Southeast Asian Conference on Geophysics, 3–5 November 2020, Bandung, Indonesia (Virtual).
- [50] J. Hestness, S. Narang, N. Ardalani, G. Diamos, H. Jun, H. Kianinejad, M. M. A. Patwary, Y. Yang, and Y. Zhou, Deep learning scaling is predictable, empirically (2017), arXiv:1712.00409 [cs.LG].
- [51] C. Sun, A. Shrivastava, S. Singh, and A. Gupta, Revisiting unreasonable effectiveness of data in deep learning era, in *2017 IEEE International Conference on Computer Vision (ICCV)* (2017) pp. 843–852.
- [52] P. Viola and M. Jones, Fast and robust classification using asymmetric adaboost and a detector cascade, *Advances in neural information processing systems* **14** (2001).
- [53] N. Bento, J. Rebelo, M. Barandas, A. V. Carreiro, A. Campagner, F. Cabitza, and H. Gamboa, Comparing handcrafted features and deep neural representations for domain generalization in human activity recognition, *Sensors* **22**, 7324 (2022).
- [54] K. Lee, K. Lee, H. Lee, and J. Shin, A simple unified framework for detecting out-of-distribution samples and adversarial attacks, *Advances in neural information processing systems* **31** (2018).
- [55] A. Torralba and A. A. Efros, Unbiased look at dataset bias, in *CVPR 2011 (IEEE, 2011)* pp. 1521–1528.
- [56] UK Parliament, Commons Library, Nuclear weapons profile: North Korea, Research Briefing CBP-9071 (2022).
- [57] W. contributors, List of nuclear weapons tests of France, https://en.wikipedia.org/wiki/List_of_nuclear_weapons_tests_of_France (2026), accessed August 2026.
- [58] UK Parliament, House of Commons, Nuclear weapons, Hansard (1990), accessed August 2026.
- [59] K. Simonyan, A. Vedaldi, and A. Zisserman, Deep inside convolutional networks: Visualising image classification models and saliency maps (2014), arXiv:1312.6034 [cs.CV].
- [60] M. Sundararajan, A. Taly, and Q. Yan, Axiomatic attribution for deep networks, in *Proceedings of the 34th International Conference on Machine Learning*, *Proceedings of Machine Learning Research*, Vol. 70, edited by D. Precup and Y. W. Teh (PMLR, 2017) pp. 3319–3328.
- [61] Y. Wang, T. Zhang, X. Guo, and Z. Shen, Gradient based feature attribution in explainable ai: A technical review (2024), arXiv:2403.10415 [cs.AI].
- [62] S. Abnar and W. Zuidema, Quantifying attention flow in transformers, in *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, edited by D. Jurafsky, J. Chai, N. Schluter, and J. Tetreault (Association for Computational Linguistics, Online, 2020) pp. 4190–4197.
- [63] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, Grad-cam: Visual explanations from deep networks via gradient-based localization, *International Journal of Computer Vision* **128**, 336–359 (2019).
- [64] J. Platt *et al.*, Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods, *Advances in large margin classifiers* **10**, 61 (1999).

Appendix A: Padding as a shortcut

Due to the rare nature and priority of the `nuclear_explosion` class, trace preservation was paramount. This is therefore, the only class that requires padding to fit the window size; padding in theory provides a perfect archival shortcut – an interpretation proving to be false. Of the 4,343 `nuclear_explosion` test traces, 3,200 (73.7%) are padded (Table IV) – padding prevalence rises across the forward-in-time split monotonically (train 42.2%, val 60.6%, test 73.7%). Test set recall, argues against shortcut: padded traces are harder to recall (0.9067) than unpadded ones (0.9685) – a shortcut would make the padded subset the easy case. Precision is higher under padding (0.9827 vs. 0.9560), but this is not cause for concern: recall is the critical metric for this class.

Pad state	n	Precision	Recall	F1
Platt, unpadded ($n_{\text{pad}} = 0$)	1,143	0.9649	0.9606	0.9627
Platt, padded ($n_{\text{pad}} > 0$)	3,200	0.9866	0.9184	0.9513
Temp, unpadded ($n_{\text{pad}} = 0$)	1,143	0.9560	0.9685	0.9622
Temp, padded ($n_{\text{pad}} > 0$)	3,200	0.9827	0.9067	0.9433

TABLE IV: Trace-level precision/recall/F1 for `nuclear_explosion` on the test split, split by whether the trace required zero-padding to fill the fixed window.

We may also verify this perceptually. A test from *Lop Nur* (China, 1994, yield 90 kt) – during the negotiation of the CTBT – shows a heavily padded trace which is correctly classified by 5 of 6 members (Fig. 8). We clearly observe spectrogram attribution mass for all methods predominantly falls on the physically meaningful region (excepting what we suspect is a small calculation artefact of attention rollout, nucleating in late time low frequency). Waveforms, as normal, seem to show saliency mass throughout the spectrum, but integrated gradients and time occlusion clearly respond well to the onset and (short) trace envelope.

We acknowledge that, perceptually, this is not always true. For transparency, we present an example where `SWP-Net` has clearly attended the padded region. This trace corresponds to *Punggye-ri* (North Korea, 2009, yield 2.4 kt) and can be observed in Fig. 9. Grad-CAM attribution mass falls largely within the padded region (although waveform attribution falls within real signal). One interpretation is that attribution is to the edge itself; the model may be concerned by the onset of any signal more so than it is with specific parts. The ViT presents something more sensible: attribution mass shows a nucleated high-low frequency split at the 20s onset. The analyst viewing this outcome may then judge the ViT as a more meaningful contributor and discard `SWP-Net` if desired – human-in-the-loop workflow.

We leave it up to the reader to further prove to themselves that attributions are generally physical. Use `SeismicEnsemble/scripts/plot_case_study_full.py`

to plot your own case studies, referring to the `README.md` for instructions.

Appendix B: Hyperparameter sensitivity

Full grids underlying Sec. IIIB (18 sweeps, 68 grid-points total – Tab. V and Tab. VI). Each hyperparameter is swept independently with every other parameter held at its tuned (Optuna-selected) value – reported scores are validation macro-F1.

Appendix C: Spatial generalisation

Full data underlying Sec. IIID and Table III: the complete 3×3 transfer matrix per model (Table VII), sector split sizes (Table VIII), and per-sector class support (Table IX).

Appendix D: Hand-crafted feature set

The 17 features below are the same set consumed by both `XGBoost` and `SVM` (`configs/models/li.yaml`, `configs/models/pignatelli.yaml`, both `model.features`). The `Phys-CNN/SWP-Net` physics branches consume an identical 4-feature subset of these – `hf_1f_energy_ratio`, `dominant_frequency`, `spectral_centroid`, `spectral_flux` (`configs/models/kong.yaml`, `configs/models/swaine.yaml`, both `model.physics_features`) – items 13, 8, 9 and 14 below. Every formula is implemented in `SeismicEnsemble/seismic_ensemble/data/features.py`. Every formula is implemented in `SeismicEnsemble/seismic_ensemble/data/features.py`. w_i is the processed (z-scored) waveform, $i = 1, \dots, N$ samples at $f_s = 20$ sps; $P_k = |\text{FFT}(w)_k|^2$ is the power spectral density at bin frequency f_k .

1. `rms_amplitude` – overall signal power.

$$\sqrt{\frac{1}{N} \sum_i w_i^2}$$

2. `peak_amplitude` – largest amplitude excursion.

$$\max_i |w_i|$$

3. `zero_crossing_rate` – how often the trace crosses zero, a coarse frequency proxy.

```
import numpy as np
out["zero_crossing_rate"] = np.mean(np.abs(
    np.diff(np.sign(w))) > 0)
```

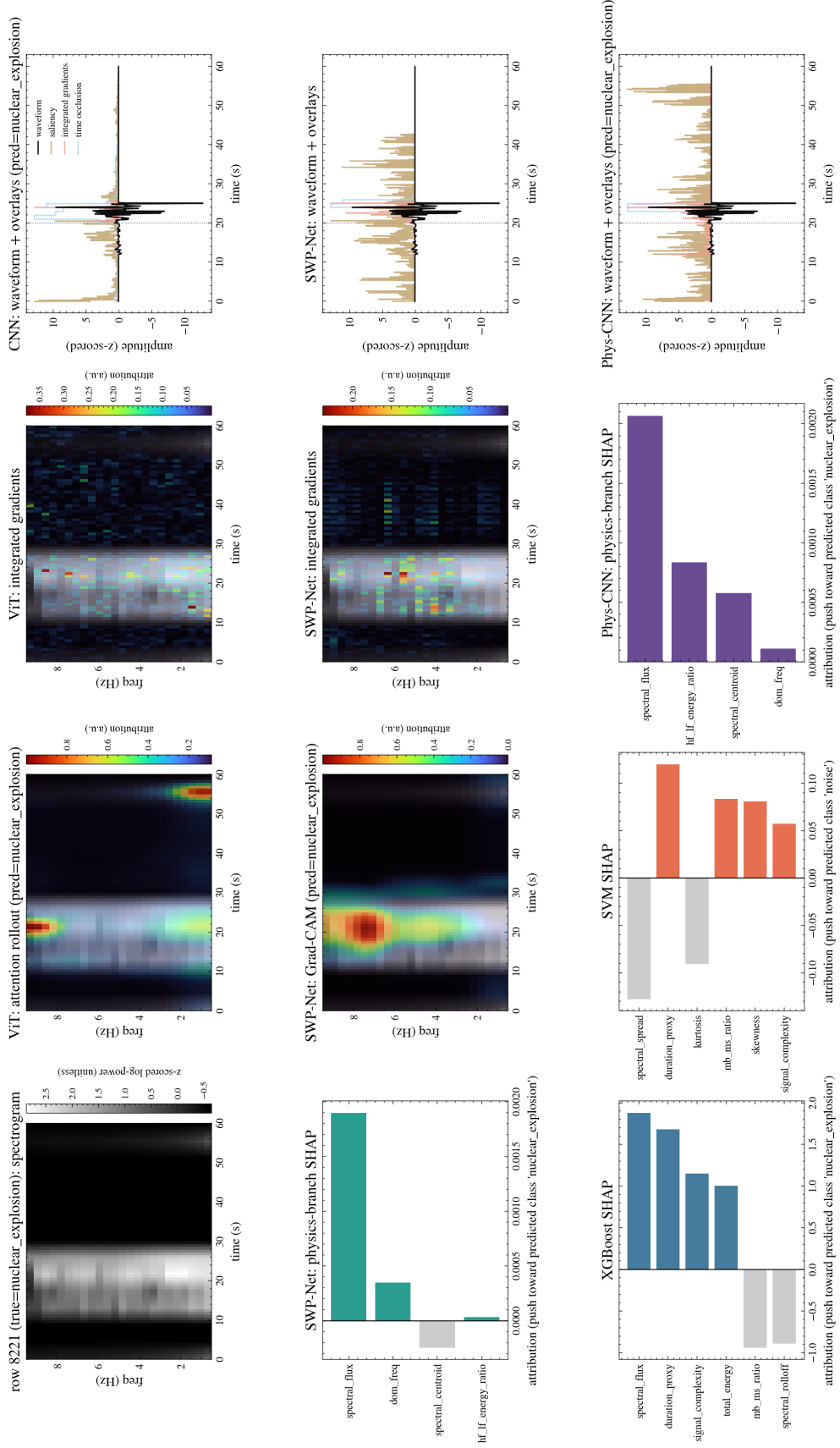


FIG. 8: Full six-member attribution on a heavily padded trace. 937 samples here are zero-padded ($\sim 78\%$). Five out of six members correctly classify this trace.

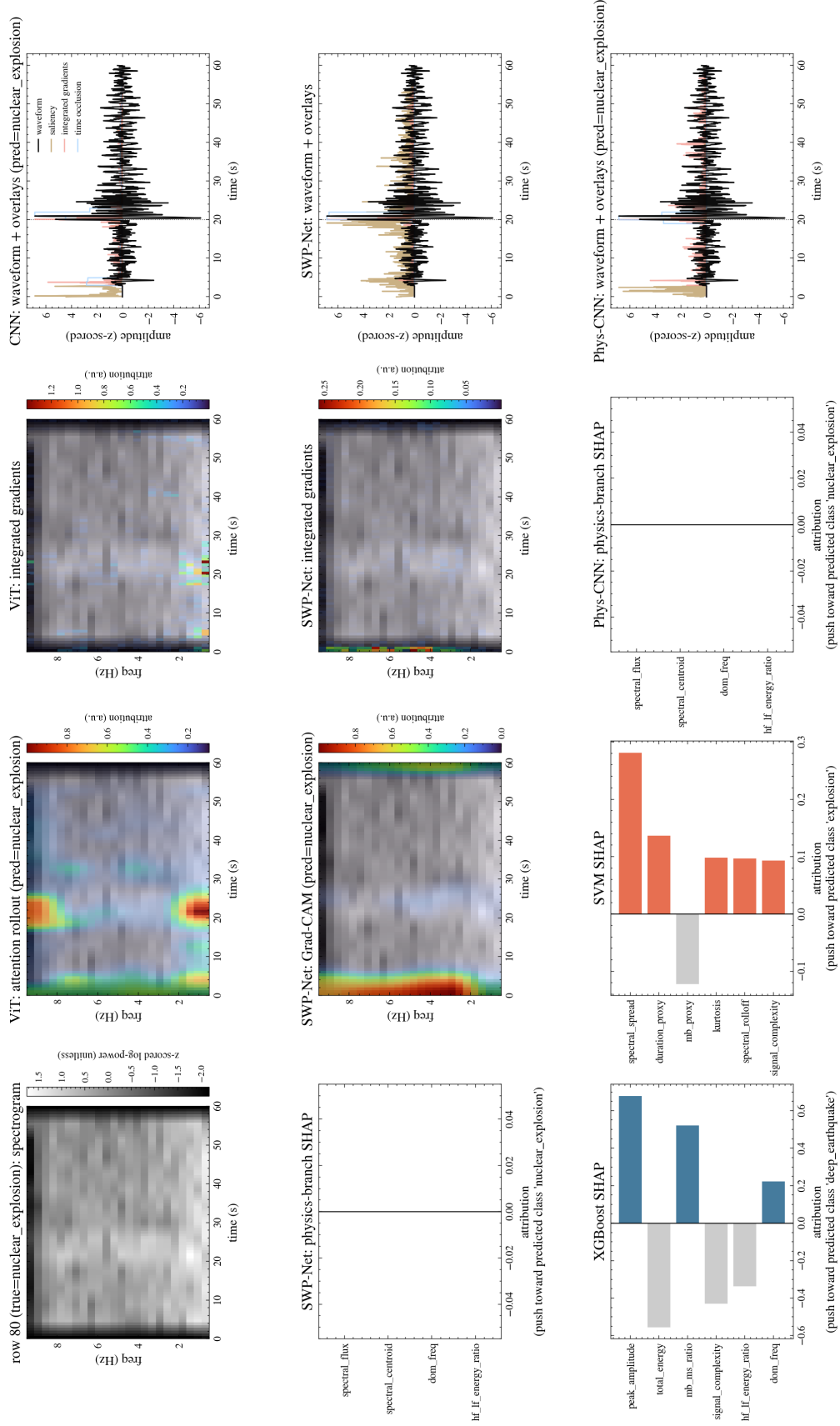


FIG. 9: Full six-member attribution on a padded trace. 59 samples here are zero-padded ($\sim 5\%$). Four of six members correctly classify this trace.

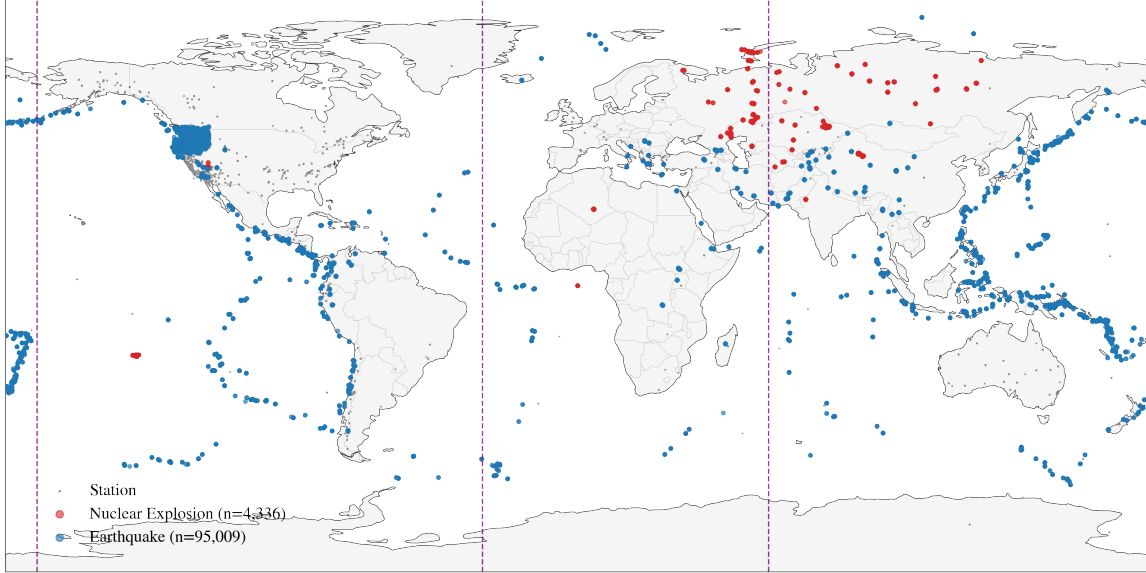


FIG. 10: Spatial experiment training split. Noise is not plotted as it has no source.

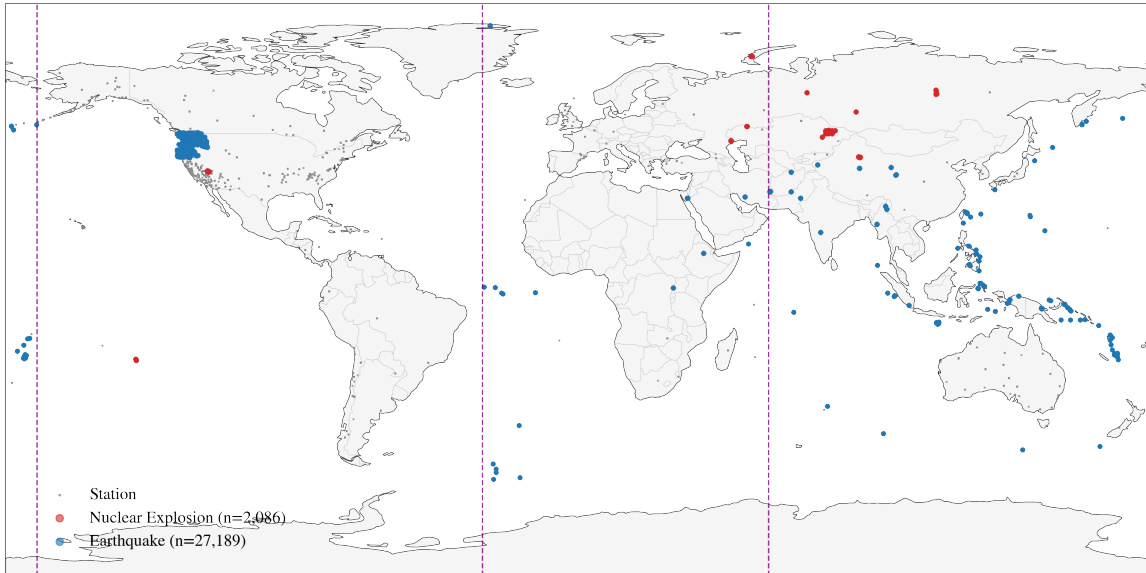


FIG. 11: Spatial experiment validation split. Noise is not plotted as it has no source.

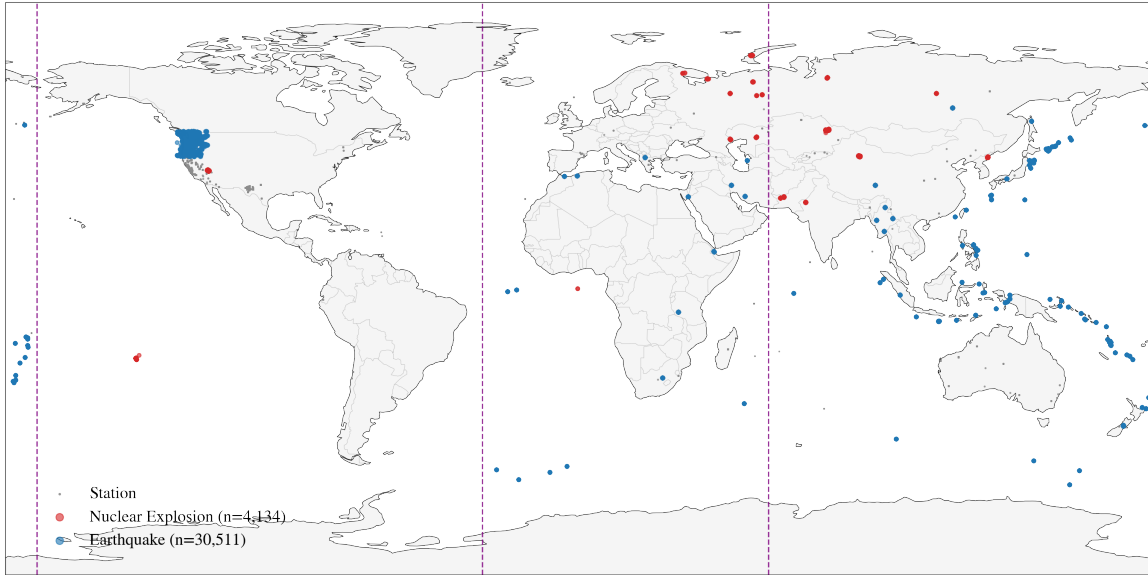


FIG. 12: Spatial experiment test split. Noise is not plotted as it has no source.

4. (Excess) **kurtosis** – tailedness of the amplitude distribution; flags a spike against a quiet trace. Subtracting 3 makes the perfect normal distribution zero. w is already z-scored.

$$\frac{1}{N} \sum_i w_i^4 - 3$$

5. **skewness** – asymmetry of the amplitude distribution.

$$\frac{1}{N} \sum_i w_i^3$$

6. **total_energy** – total energy in the window.

$$\sum_i w_i^2$$

7. **duration_proxy** – seconds spent above 20% of peak amplitude, a duration estimate.

```
env = np.abs(w)
thresh = 0.2 * (env.max() + 1e-12)
out["duration_proxy"] = np.mean(env >
    thresh) * (n / sr)
```

8. **dominant_frequency** – frequency carrying the most spectral power.

$$\arg \max_{f_k} P_k$$

9. **spectral_centroid** – power-weighted mean frequency (“brightness”).

$$\frac{\sum_k f_k P_k}{\sum_k P_k}$$

10. **spectral_spread** – bandwidth of the spectrum around the centroid.

$$\sqrt{\frac{\sum_k (f_k - f_c)^2 P_k}{\sum_k P_k}}$$

11. **signal_complexity** – normalised spectral entropy over the $K = N/2 + 1$ frequency bins; near 0 means energy concentrated in a few bins, near 1 means energy spread broadly.

$$\hat{P}_k = \frac{P_k}{\sum_j P_j}, \quad \frac{-\sum_k \hat{P}_k \ln \hat{P}_k}{\ln K}$$

12. **spectral_rolloff** – frequency below which 85% of spectral energy lies.

```
cumulative = np.cumsum(P)
roll_idx = np.searchsorted(cumulative, 0.85
    * cumulative[-1])
out["spectral_rolloff"] = freqs[min(roll_idx,
    len(freqs) - 1)]
```

13. **hf_lf_energy_ratio** – high- vs low-frequency energy split at 3 Hz; explosions tend to be HF-rich relative to earthquakes.[2]

$$\frac{\sum_{f_k \geq 3\text{Hz}} P_k}{\sum_{f_k < 3\text{Hz}} P_k}$$

14. **spectral_flux** – how much the spectral shape changes across the window. $M_j = |\text{FFT}(w^{(j)})|$ is the magnitude spectrum of the j -th of 6 equal frames; $\hat{M}_j = M_j / \sum_k M_j[k]$ is its magnitude-normalised (L2) form.

$$\frac{1}{5} \sum_{j=1}^5 \|\hat{M}_{j+1} - \hat{M}_j\|_2$$

15. **mb_proxy** – a proxy for the body-wave magnitude m_b . Log-amplitude of the trace’s energy in a (1–5 Hz) band. **band_peak**(lo, hi): zero every FFT bin outside $[lo, hi]$ Hz, inverse-transform back to the time domain, then take the peak absolute amplitude of that band-passed copy of w (a time-domain peak, not a frequency-domain magnitude) – used again below.

$$\log_{10}(\text{band_peak}(1, 5\text{ Hz}))$$

16. **ms_proxy** – a proxy for the surface-wave magnitude M_s . Log-amplitude of the trace’s energy in a (0.5–2 Hz) range.

$$\log_{10}(\text{band_peak}(0.5, 2\text{ Hz}))$$

17. **mb_ms_ratio** – the classical m_b/M_s discriminant, computed from proxies rather than catalogued magnitudes.

$$\text{mb_proxy} / \text{ms_proxy} \quad \text{clipped to } \pm 100$$

Appendix E: Attribution methods

Gradient methods evaluate how the predicted class score would move if the input moved. Plain saliency takes the magnitude of that gradient directly. Integrated gradients instead accumulates gradients along a path from a baseline (zero-tensor) to the actual trace. Grad-CAM weights a convolutional layer’s activation feature maps by their gradients and sums them. It also requires a spatial feature map to hook, which is why it is unavailable for ViT, which collapses to patch-embeddings.

Perturbation/occlusion methods mask a region and measure how far the class probability falls. In occlusion (applied here over time windows), a large drop means the model relied on that information. Its weaknesses are cost, and saturation.

Attention rollout propagates the transformer’s attention tensors across layers, accounting for residual connections, and shows where information was routed from.

SHAP assigns each input feature its average marginal contribution across coalitions of the others, giving signed attributions for the classical model family, the hybrid physics branches, and the meta-learner.

elsewhere (Tab. XIV). **Hand+SSL** is therefore the deployed configuration due to general superiority.

Appendix F: Self-supervised learning

The screening cascade, is evaluated on 20,000 test traces (a single uniform random sample over the full test split not stratified by class: 9,761 noise, 7,516 earthquake, 1,170 nuclear_explosion, 837 deep_earthquake, 674 explosion and 42 volcanic_eruption)

Appendix G: Optuna

Hyperparameters were tuned with Optuna’s tree-structured Parzen estimator (TPE), using default Optuna with a variable, trial count per model (expense adjusted). At each step, the top decile of trials by validation performance and the remaining trials are each modelled as a Gaussian kernel density estimate. The next candidate is proposed by picking one of the top-decile trials at random and perturbing slightly. This concentrates search near clusters of good trials while still introducing variation. Tab. XI

Appendix H: In-distribution performance

Appendix I: Platt Calibration

Platt calibration fits a sigmoid on each member’s validation outputs against true labels in that set, yielding calibrated probabilities before stacking with the meta-learner.[64]

Appendix J: Meta-learner feature ablation

Table XIV isolates the wave-feature block and its accepted inputs. 256 self-supervised features alone trail six hand-crafted features by 0.0063 macro-F1 at the trace level, against the 0.0104 four-seed-refit floor (not distinguishable from noise). Combining both blocks is the configuration of this work and leads on trace-level and most columns, with a margin over hand-crafted alone of 0.0200. At the event level (non-singleton), the ranking changes: **SSL only** leads (0.9626), ahead of **Hand + SSL** (0.9450) and **Hand-crafted** (0.9390) – the best configuration therefore depends on if per-trace or per-event is desired.

We also report results for temperature calibration under this regime (Tab. XV). Under temperature calibration, **Hand+SSL** leads on every metric (Tab. XV); under Platt it trails **SSL only** at the event level while leading

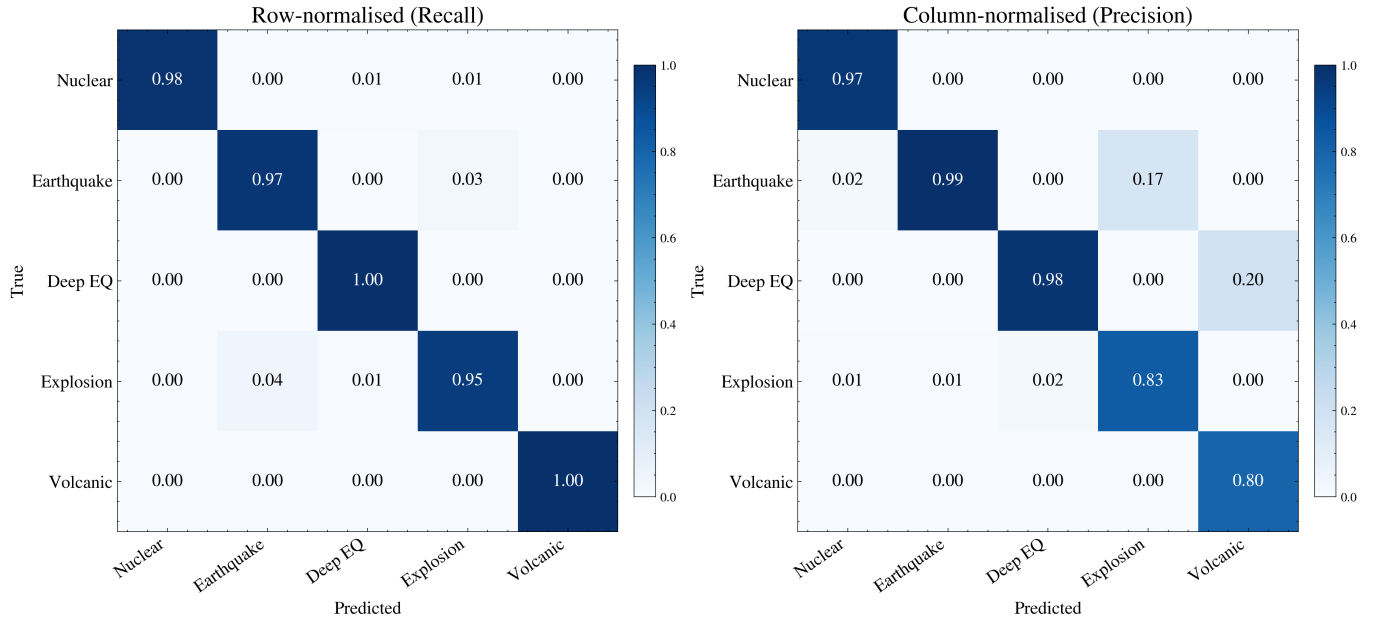


FIG. 13: Confusion matrices denoting recall and precision of event-level ($n_{\text{trace}} > 1$) stacked ensemble (meta-learner) using **Platt-calibration** predictions.

TABLE V: Hyperparameter sensitivity grids, four deep-learning members. Every value tested is listed with its resulting validation macro-F1.

Hyperparameter	Value	Val. macro-F1
CNN		
dropout	0.0	0.8385
dropout	0.1	0.6964
dropout	0.2	0.6507
dropout	0.35	0.3552
dropout	0.5	0.0914
lr	1×10^{-4}	0.7729
lr	3×10^{-4}	0.8032
lr	1×10^{-3}	0.8324
lr	3×10^{-3}	0.7995
weight_decay	1×10^{-6}	0.8450
weight_decay	1×10^{-5}	0.8426
weight_decay	1×10^{-4}	0.8386
weight_decay	1×10^{-3}	0.7876
Phys-CNN		
lr	1×10^{-4}	0.8628
lr	3×10^{-4}	0.8696
lr	1×10^{-3}	0.8547
lr	3×10^{-3}	0.8516
head_dropout	0.1	0.8829
head_dropout	0.3	0.8579
head_dropout	0.5	0.8702
head_dropout	0.7	0.8661
n_conv_layers	2	0.8653
n_conv_layers	3	0.8777
n_conv_layers	4	0.8691
ViT		
lr	1×10^{-4}	0.8264
lr	3×10^{-4}	0.8242
lr	6×10^{-4}	0.8176
lr	1×10^{-3}	0.8265
dropout	0.0	0.8164
dropout	0.1	0.8347
dropout	0.2	0.8401
dropout	0.3	0.8170
embed_dim	128	0.8426
embed_dim	192	0.8345
embed_dim	256	0.8267
SWP-Net		
n_cnn_blocks	2	0.8733
n_cnn_blocks	3	0.8943
depth	1	0.8958
depth	2	0.8384
depth	3	0.8911
head_dropout	0.1	0.9001
head_dropout	0.3	0.8882
head_dropout	0.5	0.8905
head_dropout	0.7	0.8502
wave_n_conv_layers	2	0.8378
wave_n_conv_layers	3	0.9002
wave_n_conv_layers	4	0.8860

TABLE VI: Hyperparameter sensitivity grids, two classical members. Every value tested is listed with its resulting validation macro-F1.

Hyperparameter	Value	Val. macro-F1
XGBoost		
n_estimators	200	0.6214
n_estimators	600	0.6623
n_estimators	1200	0.6816
n_estimators	2000	0.6864
learning_rate	0.01	0.6054
learning_rate	0.05	0.6623
learning_rate	0.1	0.6822
learning_rate	0.3	0.6788
max_depth	3	0.6035
max_depth	6	0.6623
max_depth	9	0.6873
max_depth	12	0.6912
SVM		
C	0.1	0.5179
C	1	0.5685
C	10	0.5920
C	100	0.6036
C	1000	0.6340
gamma	1×10^{-3}	0.5251
gamma	0.01	0.5505
gamma	0.1	0.5965
gamma	1	0.6109

TABLE VII: Spatial transfer, full 3×3 macro-F1 matrices (evaluatable classes only: **nuclear_explosion**, **earthquake**, **noise**), all six members. Rows are the training sector, columns the test sector.

Train sector	Americas	EMEA	AsiaPac
CNN			
Americas	0.9388	0.8438	0.7994
EMEA	0.3159	0.9150	0.8405
AsiaPac	0.3610	0.9019	0.8658
Phys-CNN			
Americas	0.9390	0.7857	0.7627
EMEA	0.3419	0.8915	0.8841
AsiaPac	0.3799	0.9094	0.9189
ViT			
Americas	0.9572	0.8760	0.8412
EMEA	0.3282	0.8966	0.8149
AsiaPac	0.3348	0.9251	0.9093
SWP-Net			
Americas	0.9471	0.8878	0.8611
EMEA	0.3301	0.8935	0.8978
AsiaPac	0.3506	0.9247	0.8875
XGBoost			
Americas	0.8421	0.7064	0.5922
EMEA	0.3426	0.7108	0.6716
AsiaPac	0.3730	0.7936	0.8088
SVM			
Americas	0.8393	0.6724	0.5410
EMEA	0.4109	0.8597	0.7920
AsiaPac	0.3586	0.7834	0.7286

TABLE VIII: Sector split sizes (traces, all six classes).

Sector	Train	Val	Test
Americas	240,420	57,393	60,366
EMEA	11,858	2,787	2,896
AsiaPac	38,477	10,024	12,452
Total	290,755	70,204	75,714

TABLE IX: Per-sector trace counts, train, val and test splits, all six classes. 30 traces minimum: `nuclear_explosion`, `earthquake` and `noise` (a 30 trace count was motivated by experimental pragmatism; it does not have distinct physical reasoning).

Sector	Split	Nuclear	Earthquake	Deep EQ	Explosion	Noise	Volcanic
Americas	Train	1,180	86,863	783	9,920	140,861	813
Americas	Val	692	23,607	145	2,649	30,185	115
Americas	Test	899	26,533	131	2,528	30,185	90
EMEA	Train	952	1,004	56	11	9,835	0
EMEA	Val	355	318	6	1	2,107	0
EMEA	Test	317	457	13	2	2,107	0
AsiaPac	Train	2,204	7,142	14,931	42	14,102	56
AsiaPac	Val	1,039	3,264	2,693	6	3,022	0
AsiaPac	Test	2,919	3,521	2,982	8	3,022	0
Total		10,557	152,709	21,740	15,167	235,426	1,074

TABLE X: Attribution methods used, by model and representation. “Signed” indicates whether the output distinguishes evidence for the predicted class from evidence against it; unsigned maps show magnitude only. Resolution is the granularity of the returned map, not of the input.

Method	Family	Signed	Resolution
Saliency	gradient	no	per sample
Integrated gradients	gradient	yes	per sample / bin
Grad-CAM	gradient	no	conv feature map
Time occlusion	perturbation	yes	window
Attention rollout	attention	no	patch
SHAP	additive	yes	per feature

TABLE XI: Best Optuna-tuned hyperparameter configuration per member. After 20 tested configurations per model.

Model	Tuned hyperparameters
CNN	dropout=0.0172, lr= 3.5068×10^{-3} , weight_decay= 5.9750×10^{-6} , batch_size=128
ViT	embed_dim=192, depth=6, dropout=0.1777, attn_dropout=0.0093, lr= 4.0508×10^{-4} , weight_decay= 2.1930×10^{-3}
Phys-CNN	cnn_dropout=0.3860, head_dropout=0.3075, root_filters=96, n_conv_layers=2, lr= 5.5108×10^{-4} , weight_decay= 1.3048×10^{-5}
SWP-Net	root_channels=32, n_cnn_blocks=3, cnn_dropout=0.1087, depth=3, dropout=0.2250, attn_dropout=0.1939, head_dropout=0.2975, physics_hidden=64, lr= 2.6511×10^{-4} , weight_decay= 5.8251×10^{-4} , wave_root_filters=96, wave_n_conv_layers=3, wave_dropout=0.0926, wave_embed_dim=128
XGBoost	n_estimators=1900, max_depth=8, learning_rate=0.0850, subsample=0.6278, colsample_bytree=0.5060, min_child_weight=2.0585, gamma=0.1809, reg_lambda=0.0304, reg_alpha=0.0100
SVM	kernel=rbf, C=44.7671, gamma=0.1983

TABLE XII: Test-split performance under Platt calibration, provided for comparison against Table II (temperature calibration, main text). Member values are as-trained (uncalibrated) and identical to Table II; the ensemble consumes Platt-calibrated probabilities, whose per-member macro-F1 is CNN 0.8462, Phys-CNN 0.8766, ViT 0.8372, SVM 0.5906, XGBoost 0.6274, SWP-Net 0.8780, alongside hand-crafted and self-learned features. Unlike temperature scaling, Platt calibration is not constrained to preserve each member’s predicted class, so these calibrated figures differ from the as-trained values above. The direct comparison between ensemble and individual member is therefore 0.8974 against SWP-Net’s 0.8780. The best individual member output per column is emboldened. F1 on **noise** is omitted due to near-complete resolution. Meta-learner outputs are displayed at the trace and event-level. Individual model outcomes are reported at the trace-level. NLL is negative log-likelihood.

Model	Acc. $\uparrow$	Macro-F1 $\uparrow$	Bal. Acc. $\uparrow$	NLL $\downarrow$	F1 _{nuc} $\uparrow$	F1 _{eq} $\uparrow$	F1 _{deep} $\uparrow$	F1 _{exp} $\uparrow$	F1 _{volc} $\uparrow$
SWP-Net	0.9667	0.8757	0.8705	0.1872	0.8962	0.9684	0.9087	0.7643	0.7212
Phys-CNN	0.9640	0.8620	0.8739	0.1245	0.8867	0.9683	0.8883	0.7439	0.6894
ViT	0.9614	0.8412	0.8340	0.2226	0.8532	0.9657	0.8921	0.7483	0.5929
CNN	0.9364	0.7872	0.8656	0.2232	0.7617	0.9453	0.8188	0.6292	0.5749
XGBoost	0.8834	0.6108	0.6867	0.3780	0.5196	0.9229	0.6497	0.4734	0.1345
SVM	0.8905	0.5796	0.5934	0.4728	0.4896	0.9233	0.6474	0.4525	0.0000
Ensemble – Trace	0.9769	0.8974	0.9405	0.1473	0.9589	0.9767	0.9548	0.8030	0.6944
Ensemble									
Event-level ($n_{\text{trace}} > 1$)	0.9725	0.9448	0.9807	0.1139	0.9771	0.9841	0.9885	0.8854	0.8889

TABLE XIII: Per-class event and trace counts for events surviving the $n_{\text{traces}} > 1$ event-level cut (Sec. III A). **noise** has no multi-trace events by construction.

Class	Events	Traces	Traces/event
nuclear_explosion	130	4,325	33.27
earthquake	5,144	26,528	5.16
deep_earthquake	303	3,101	10.23
explosion	676	2,229	3.30
noise	0	0	–
volcanic_eruption	4	146	36.50
Total	6,257	36,329	–

TABLE XIV: Meta-learner feature ablation (**Platt-calibrated**). All three ensembles are identical except for the feature block used. SSL features are a 256-d concatenation of two frozen masked-autoencoder encoders (waveform and spectrogram, 128 dimensions each at the trace level). Event-level macro-F1 restricts scoring to events with more than one contributing trace and excludes **noise** (Sec. III A), retaining 6,257 of 42,728 events for all three variants.

Wave features	M-F1 _{tr}	Acc.	M-F1 _{ev}	F1 _{nuc}	F1 _{exp}
Hand-crafted (6)	0.8774	0.9708	0.9390	0.9374	0.7708
Hand+SSL (262)	0.8974	0.9769	0.9450	0.9589	0.8030
SSL only (256)	0.8711	0.9691	0.9626	0.9336	0.7560

TABLE XV: Meta-learner feature ablation, **temperature-calibrated**. All three ensembles are identical except for the feature block used. SSL features are a 256-d concatenation of two frozen masked-autoencoder encoders (waveform and spectrogram, 128 dimensions each at the trace level). The 17 hand-crafted features are listed in App. D. Event-level macro-F1 restricts scoring to events with more than one contributing trace and excludes **noise** (Sec. III A), retaining 6,257 of 42,728 events for all three variants.

Wave features	M-F1 _{tr}	Acc.	M-F1 _{ev}	F1 _{nuc}	F1 _{exp}
Hand-crafted (6)	0.8904	0.9699	0.9370	0.9329	0.7574
Hand+SSL (262)	0.9104	0.9770	0.9451	0.9542	0.7988
SSL only (256)	0.8689	0.9670	0.9382	0.9208	0.7537

Appendix K: Uncertainty features

Alongside the calibrated class-probability vectors (6×6), the meta-learner receives 8 uncertainty features (`seismic_ensemble/ensembling/meta_learner.py`, `stack_features`; `aggregator.py` for disagreement).

1. **disagreement** – mean pairwise total-variation distance between members’ probability vectors.

```
for i in range(M):
    for j in range(i + 1, M):
        total += 0.5 * np.abs(P[i] - P[j]).
    sum(axis=1)
    pairs += 1
disagreement = total / pairs
```

2. **entropy** – Shannon entropy of the members’ averaged probability vector.

$$\bar{p} = \frac{1}{M} \sum_m p_m, \quad - \sum_c \bar{p}_c \ln \bar{p}_c$$

3. **confidence** – one column per member, that member’s own top-class probability (6 columns total).

$$\max_c (p_m)_c$$

Appendix L: P-wave estimation method

Per-trace P-wave duration (Sec. IIIF) uses the smoothed Hilbert-envelope amplitude. The Hilbert transform is the linear time invariant (LTI) system $\mathcal{H}[w](t) = (w * h)(t)$ with impulse response $h(t) = 1/(\pi t)$, whose frequency response $\hat{h}(f) = -i \operatorname{sgn}(f)$ is a flat 90° phase shift (magnitude unchanged). This gives the envelope a smooth amplitude curve, unlike on the raw oscillating waveform. After 0.5s (10 sample moving average window) smoothing, pre-pick median sets the noise floor, the post-pick (within 40s) peak marks the P-wave, and duration goes from the peak to halfway back to the floor. Implemented in `seismic_ensemble/experiments/pick_perturbation.py`, `estimate_p_duration_s`.

```
env = np.abs(hilbert(wf))
env = np.convolve(env, np.ones(k) / k, mode
    ="same") # k = 0.5 s smoothing
floor = np.median(env[:pick_sample])
post = env[pick_sample : pick_sample + int
    (40 * sps)]
peak_i, peak = np.argmax(post), post.max()
thr = floor + 0.5 * (peak - floor)
```

```
duration_s = (peak_i + np.where(post[peak_i
:] < thr)[0][0]) / sps
```

Falls back to NaN (and the class median elsewhere, Sec. IIIF) if no discernible post-pick energy is found ($\text{peak} \leq 1.5 \times \text{floor}$) or the pick sits too close to either edge of the trace.

Appendix M: Standard error

Table II and similar tables report to four decimal places. For a simple proportion $\hat{p}$ over n trials, $\text{SE}(\hat{p}) = \sqrt{\hat{p}(1 - \hat{p})/n}$. Worked example, trace-level accuracy ($\hat{p} = 0.9770$, $n = 72,800$):

$$\text{SE} = \sqrt{\frac{0.9770 \times 0.0230}{72,800}} \approx 0.00056$$

i.e. ± 0.06 percentage points – smaller than a single unit change in the third decimal place. This formula assumes independent trials; many traces here share the same physical event across stations, so it understates the true standard error somewhat – the value above is a floor, not an exact figure.

The standard error matters here for a specific reason: configurations evaluated on the same test set are resolved more precisely than either figure’s own marginal standard error suggests, since sampling noise shared by both configurations cancels in the delta. That is the motivation for retaining this precision throughout the report.

Appendix N: Average traces per event

Average traces per training event: `noise` 1.0, `explosion` 2.2, `earthquake` 3.6, `nuclear_explosion` 8.5, `deep_earthquake` 10.6, `volcanic_eruption` 47.1 (`volcanic_eruption` 754 traces over 16 events). Percentage of events still present at 10% traces remaining: 10.0, 19.9, 28.5, 45.1, 58.6 and 100.0 respectively. We note that `volcanic_eruption` has significant data-volume per event despite few events, `volcanic_eruption` survives by fortune under the fixed seed – not as a structural property.

Appendix O: Depth as a leakage diagnostic

Two additional variants: depth as returned from the meta-data, imputing 0 where depth is unavailable (C) and B repeated with `nuclear_explosion`’s unknown flag reassigned to shallow (0km), denoted (D). Members, calibration and splits were held fixed. Adding depth improves trace macro-F1 substantially: 0.8774 (A) to 0.9428 (B), 0.9056 (C) and 0.9225 (D), with nuclear F1 rising from 0.9374 to 0.9829 under B and deep-earthquake F1 reaching 1.0000 under C. Across four seeds the A and B ranges do not overlap, so the effect is not noise.